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Design and Development of a Tangible Learning System for Special Education

submitted by
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Declaration

The author of this thesis used the artificial intelligence tool ChatGPT to support the writing process by improving the English wording and linguistic structure of the thesis. All content generated or suggested by artificial intelligence tools was carefully reviewed, revised, and verified by the author.

I hereby declare that I have written this thesis independently and have used no sources or aids other than those indicated.
Augsburg, 18 August 2026

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Abstract

This thesis …

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1 Special Education Context

Students with special educational needs in the areas of Special education focus on intellectual development (**GE**) (*German: Förderschwerpunkt Geistige Entwicklung, GE*), Special education focus on learning (**LE**) (*German: Förderschwerpunkt Lernen, LE*) and Autism Spectrum Disorder (**ASD**) may face a common range of challenges in the learning process. These may include difficulties with fundamental prerequisites for learning, such as perseverance, memory and motivation, as well as with learning how to learn. Learners may experience difficulties reflecting on their own learning, evaluating their learning processes and transferring acquired knowledge and skills to everyday situations. Challenges related to independent thinking, action planning and problem-solving may also be relevant, as these abilities enable learners to structure sequences of actions and cope with emerging problems independently [1–3].

Against the background of these challenges, the following areas of development and competence are particularly relevant [1–4]:

- Action planning
- Problem-solving
- Recognition of relationships
- Attention and memory
- Motor skills and perception
- Independence and self-regulation

Special educational provision therefore aims to support students in developing these competencies, enabling them to act with increasing independence and self-determination in both educational and everyday contexts and to participate more fully in society. These competencies extend beyond the possession of knowledge and include the ability to make decisions, plan and carry out actions, evaluate their outcomes and seek appropriate support when required [1, 4, 5].

1.1 Implications for Learning and Teaching

As shown in Table 1, the different difficulties and needs, and consequently the heterogeneity of learner groups resulting from the three areas considered, **GE**, **LE** and **ASD**, make it difficult to select educational opportunities that are equally suitable for all learners [5].

Area	Description	Support Approach
GE	Varying levels of cognitive development, motor skills and perception	Concrete, action-oriented approaches, individual support and sufficient processing time
LE	Difficulties with academic learning such as learning strategies and the transfer of acquired knowledge	Clear subtasks, repetition and comprehensible feedback
ASD	Individually varying characteristics in perception, communication and interaction	Clear structures, visualised processes and individually adapted levels of sensory stimulation

Table 1: *Relevant developmental areas in special education and corresponding support approaches [1–3]*

In the area of learning, for example, the acquisition of basic academic skills and their transfer to other situations represents a central educational objective [2], whereas students in the area of intellectual development may require a more development-oriented form of support that addresses fundamental cognitive competencies such as cognition, perception and motor skills [1]. At the same time, students within each of the three areas do not form a homogeneous group and therefore require individually adapted support. This is particularly evident in autism spectrum disorder, as

each child presents a unique combination of characteristics. The disorder may occur in conjunction with severe cognitive impairments, but also with high cognitive abilities [3]. This heterogeneity therefore requires an inclusive educational approach that builds on the individual prerequisites of each child and provides appropriately adapted learning opportunities and support measures.

Despite these differences, several common principles for the design of suitable learning opportunities can be derived across the three areas [1–3]:

- Individual adaptation rather than standardised requirements
- Clear and manageable task structures
- Visualisation of processes and sequences of actions
- Sufficient processing time and opportunities for repetition
- Playful and action-oriented tasks
- Timely and comprehensible feedback
- Promotion of independence and self-efficacy

One way of addressing different learning prerequisites is to make learning content accessible through different perceptual and action-based approaches. In this context, the interaction between motor and perceptual experience plays a particularly important role, since learners can develop sensorimotor integration processes and support learning at an elementary level through active engagement and the conscious perception of their own movements and environment [1].

A suitable framework for such action-oriented experiences is provided by playful and constructive forms of learning, as they offer opportunities to develop and strengthen sensory, motor and cognitive abilities within concrete contexts of action. At the same time, they enable exploratory experimentation in which errors are accepted as part of the learning process [6].

In addition to designing appropriate learning approaches, the promotion of self-directed learning represents another important educational objective intended to strengthen students' independence [7]. However, the ability to complete a task independently requires learners to recognise the effects of their actions and adapt their subsequent approach accordingly. Timely and comprehensible feedback is therefore essential, because otherwise there is a risk that incorrect procedures are continued unnoticed or uncertainty arises regarding how to proceed with the task. In heterogeneous or inclusive classrooms, however, a teacher cannot continuously observe every action of every student and provide immediate feedback. Particularly during phases of independent practice, a delay may therefore occur between an action and the corresponding feedback. For learners who depend on clear confirmation or correction in order to continue a task, this can make independent task completion more difficult [8].

1.2 Existing Learning Materials

Construction materials such as building blocks are generally well suited to action-oriented and self-directed forms of learning because they combine concrete actions with the exploration of spatial and logical relationships. Through active construction, experimentation and problem-solving, children can engage with learning content in a playful manner. At the same time, the result remains visible as a physical artefact and can be directly examined and modified. However, construction materials are generally unable to independently detect or evaluate the resulting construction. The feedback required for self-directed learning must therefore be provided either by the learners themselves or by a teacher [9].

At the same time, digital technologies such as tablets make it possible to present learning content through multimedia, adapt tasks to individual needs and provide automated feedback. However, spatial content is often represented only on a two-dimensional screen, while interaction is largely limited to mouse or touch input. Although these forms of input also enable active interaction, they do not provide the same haptic and spatially differentiated experience as the

manipulation of three-dimensional objects. This reduces the physical and motor interaction with learning objects, which is highly relevant for successful learning, to a minimum [10].

1.3 Objectives and Scope

In this context, the objective of this thesis is to develop a modular, brick-based learning game that combines the haptic and spatial properties of conventional construction materials with the possibilities of digital detection, visualisation and feedback. Within a guided construction task, children can use the bricks to build a three-dimensional structure and receive immediate feedback on whether their construction corresponds to the target structure or contains errors. Feedback is provided both directly through the bricks and through an additional visualisation on a display. Power is supplied centrally by the baseplate and distributed through all connected bricks.

The physical manipulation of the bricks combines visual, haptic and motor experiences, while the digital representation provides additional information about the detected state of the construction. Through immediate feedback, the learning game is intended to enable learners to independently monitor and correct individual action steps. In this way, it aims to support self-directed learning, action planning, spatial and relational thinking and problem-solving.

The thesis is limited to the development and validation of a functional prototype. The size and complexity of supported structures are constrained by the available bricks and the system architecture. Product maturity, large-scale manufacturing and a dedicated tablet application for mobile devices are outside the scope of this work and remain subjects for future development.

Figure 1: Final prototype of the brick-based learning game

1.3.1 Structure

To that end, the theoretical foundations of learning and tangible user interfaces are first examined, followed by a review of existing brick-based Tangible User Interface (**TUI**) systems. Based on the resulting pedagogical and technical insights, the requirements for the learning system are derived and translated into a coherent system concept. The subsequent technical realisation describes the mechanical and electronic implementation of the bricks and baseplate as well as the embedded firmware and software of the learning application. Finally, the developed prototype is evaluated with regard to the previously defined requirements and the results, limitations and opportunities for future development are discussed.

2 Theoretical Background

2.1 Fundamentals of Learning Theory

The ways in which children acquire and apply new knowledge can be explained through different theoretical perspectives. These perspectives differ primarily in their understanding of how knowledge is formed, which role learners assume within the learning process and how learning environments should be designed. The three approaches discussed in the following, instructivism, constructivism and constructionism, pursue different educational objectives but are all relevant to the development of learning systems. While instructivism primarily conceptualises learners as recipients of externally structured knowledge, constructivism and constructionism emphasise their active participation in the learning process [11, 12].

2.1.1 Instructivism

Instructivism is based on the assumption that knowledge can be systematically structured and transmitted by a teacher. Learning primarily takes place through explanation and the reproduction of predefined content or procedures. The learner therefore assumes a comparatively receptive role, while the teacher determines the learning objectives, organises the subject matter and provides the intended solution path [11].

2.1.2 Constructivism

In contrast to the instructivist approach, constructivism defines learning as an active process in which learners independently engage with and shape the subject matter. Vollmers illustrates a fundamental principle of Piaget's theory of learning using the following quotation from Piaget [11, p. 7]:

> "Thinking means constructing relationships. The individual comprehends the external world only by establishing relationships between its elements."

Knowledge is therefore not conceived as a complete and objective representation of reality that exists independently of the learner. Instead, it emerges as learners establish relationships between elements of their environment, interpret these relationships and integrate them into their existing knowledge structures. The conceptual understanding of such abstract relationships is preceded by concrete experiences arising from physical interaction with objects in the environment. It is recommended that objects be touched, moved, compared with one another or arranged spatially, with a view to generating experiences that provide the basis for the development of cognitive structures [11].

Based on Piaget's theory, learning is therefore described as an "interactive process between the learner and the external world" [11, p. 75]. New knowledge does not emerge solely through the presentation of information, but primarily through active engagement with the object of learning.

2.1.3 Constructionism

The mathematician and psychologist Seymour Papert extends the constructivist approach by adding an external dimension in his constructionist theory of learning. He argues that the construction of knowledge in the mind is most effective when it is supported by the construction of a tangible object in the physical world. Following the principle of "learning by making," learners reconstruct knowledge while creating something of their own, thereby connecting the external construction of an artefact with the internal construction of knowledge [12].

The constructed artefact fulfils several functions: On the one hand, it makes abstract theoretical considerations concrete, tangible and, in the literal sense, comprehensible. Relationships that may be difficult to understand when represented solely through language or graphics can become visible through the spatial arrangement of a physical model. On the other hand, modifying the model enables learners to test different assumptions and directly observe their effects [13, 14].

Papert even goes beyond the physical act of construction by emphasising the importance of personal meaning. Following the principle that "love is a better master than duty" [13, p. 6], he argues that learning is enhanced when children create something that is personally meaningful to them.

Building on the explanations of the three learning theories, the central characteristics and differences are again summarised in Table 2:

Approach	**Guiding principle**	**Construction of knowledge**	**Learning process**	**Role of learner**	**Role of teacher**
Instructivism	*Learning by listening*	Presented and transmitted through structured instruction	Predetermined content and solution paths	Passive recipient	Transmits and explains knowledge
Constructivism	*Learning by doing*	Constructed through cognitive processes, reflection and the recognition of relationships	Exploration and problem-solving	Active participant	Guides and supports learning
Constructionism	*Learning by making*	Constructed through the creation of a tangible, personally meaningful artefact	Exploration, problem-solving and iterative construction	Active creator	Guides and supports learning

Table 2: *Comparison of learning approaches [11, 13, 14]*

2.1.4 Synthesis and Conclusion

Within the context of constructivism and constructionism, success is as important as failure and the occurrence of errors. Learning does not result merely from carrying out an activity. Instead, it develops through cognitive engagement with the constructed object and with the problems encountered during the construction process. Such difficulties create an opportunity for learners to review their planning and procedures and to reflect consciously on how the situation can be resolved and how their actions might be coordinated more effectively in future attempts [11].

Compared to instructivism, this conception of learning consequently also brings about a transformation in the role of the teacher: Rather than being responsible for transmitting knowledge in its entirety, the teacher is expected to create appropriate learning opportunities, prepare didactically structured tasks and support learners when difficulties arise without anticipating or prescribing the complete solution [14].

However, the theoretical perspectives are not necessarily mutually exclusive in practical educational settings. Instructive elements may provide essential background knowledge or initial guidance, while constructivist and constructionist activities allow learners to actively apply, examine and further develop this knowledge. The appropriate combination depends on the learning objective, the learners' prior knowledge and the degree of guidance required.

2.2 Embodied Cognition and Interaction

While constructivism and constructionism explain why actively creating an externally perceivable artefact can support learning, they do not fully state the role played by bodily action and sensory perception during the construction process. This relationship is addressed by the theory of *embodied cognition*, which assumes that cognitive processes are inseparably connected to the body. *Embodied interaction* transfers this theoretical perspective to the design of interactive digital systems.

2.2.1 Embodied Cognition

Embodied cognition assumes that cognitive processes do not occur independently of the body but are shaped by perception, movement and interaction with the environment. From this perspective, the physical manipulation of a learning object is not merely a means of executing an action that has already been fully planned mentally. Instead, the manipulation itself can form part of the processes of thinking and understanding and may therefore be considered an integral component of the cognitive system. Knowledge consequently emerges through the reciprocal relationship between the brain, the body and the environment [15, 16].

For learning processes, this implies that physical activities can be particularly relevant when they correspond meaningfully to the subject matter, because arbitrary movement alone does not necessarily improve learning. Rather, the correspondence between the physical action and the concept to be learned is crucial. When learners are expected to understand a spatial relationship, for example, it may be beneficial for them to establish this relationship by physically positioning and rotating objects. In this case, the action does not merely represent the learning content symbolically but makes it directly accessible through bodily experience [15, 16].

However, the educational value of physical interaction depends on the relationship between the performed action and the concept to be learned. Movement alone does not necessarily improve learning. The physical action must meaningfully correspond to the underlying content and make relevant properties or relationships directly perceivable [16].

2.2.2 Embodied Interaction

While *embodied cognition* provides a theoretical perspective on the relationship between the body and cognition, Paul Dourish formulated *embodied interaction* as a framework for understanding and designing interactive computer systems. Rather than viewing interaction as abstract information processing, this framework emphasises how meaning emerges through users' engaged actions with artefacts and within social practices [16].

Dourish distinguishes between two dimensions of embodiment: physical presence in the world and social embeddedness within established practices and systems of meaning. Again, an object is not embodied merely because it exists in material form; what is decisive is how it is used and interpreted within a specific situation. Similarly, a user interface does not automatically enable *embodied interaction* simply because it involves physical actions. The actions performed through the interface must be comprehensible to users and must have a recognisable relationship to the system's responses [16].

The aim of *embodied interaction* is therefore to design computer systems that do not restrict users to purely abstract forms of interaction but instead make effective use of their natural, everyday abilities and established ways of acting [16].

2.2.3 Conclusion

Taken together, these theories indicate that the educational value of a tangible learning system does not result from physical activity or digital technology alone. It emerges from the coherent relationship between the learning objective, the actions performed by learners and the physical artefacts being manipulated. A suitable learning environment should therefore combine structured guidance with opportunities for independent exploration, construction and meaningful bodily interaction.

2.3 Graphical vs Tangible User Interface

A User Interface (**UI**) can be defined as the point of contact through which people communicate and interact with a technical system. The most common type of **UI** is the Graphical User Interface (**GUI**), in which digital information is visually represented on a screen and manipulated through input devices such as a mouse, a keyboard, or a touchscreen [10].

Shaer and Hornecker contrast this mode of interaction with people's everyday experience of the physical environment [10, p. 1]:

> "We live in a complex world, filled with myriad objects, tools, toys and people. Our lives are spent in diverse interaction with this environment. Yet, for the most part, our computing takes place sitting in front of and staring at, a single glowing screen attached to an array of buttons and a mouse"

The quotation highlights the separation between traditional computer interaction and people's embodied experience of the physical world: In everyday life, objects can be grasped, combined and explored through several senses, whereas conventional **GUI** translate these actions into interactions with visual representations on a two-dimensional display. Consequently, digital interaction frequently makes only limited use of people's natural physical capabilities, potentially resulting in a reduced level of intuitiveness [10].

The concept of Ishii and Ullmer's Tangible Bits responds to this separation by coupling digital information with everyday physical and graspable objects and the surrounding environment. Its central aim is to bridge the divide between the digital domain of bits and the physical world of atoms, thereby making digital information directly accessible through physical manipulation. Tangible Bits consequently provides an important conceptual foundation for the development of **TUI** [17, 18].

Unlike a **GUI**, a **TUI** integrates computing into the physical environment. Interaction does not occur exclusively through graphical elements on a screen; instead, users manipulate real-world objects, commonly referred to as phicons, to interact with the digital system. These phicons do not merely serve as input devices but also fulfil a representational function [10].

The fundamental principle of a **TUI** is predicated on two closely related characteristics [19]:

- Phicons are used as representations of digital information and computational operations
- The physical manipulation of these phicons is used to interact with the associated computational operation

Because different types of information or functions can be represented by separate physical elements, multiple phicons may remain available for simultaneous manipulation. They may also be operated in parallel or with both hands. This principle, known as *space multiplexing*, can improve

interaction efficiency and facilitate collaborative use by groups, which is typically more restricted in conventional GUI-based interaction [19].

2.3.1 MVC and MCRpd Model

The difference between these styles of user interfaces becomes more obvious when comparing their underlying interaction models shown in Figure 2 [20].

MVC model for GUI

Within the Model View Controller (**MVC**) architecture, an interactive software system is divided into three functional components: The *model* represents the underlying data and the current state of the system, the *view* presents this information graphically, while the *control* processes user input and translates it into changes to the model. Therefore, in the interaction model of a **GUI**, there is a clear separation between the control devices, such as the mouse and keyboard, and the digital representation displayed on the screen [20].

MCRpd model for TUI

The interaction model for a **TUI** retains the **MVC** *model* and *control* components but replaces the exclusively graphical *view* with both a Physical Representation (**rep_p**) and a Digital Representation (**rep_d**). It thereby accounts for the fact that, in a **TUI**, digital information is not represented solely on a screen but is also given a tangible physical form. The structure of the physical components can be characterised by three central properties:

1. The physical representation is coupled with digital information
2. The physical representation incorporates mechanisms for interactive control
3. The physical state of the object contains information about the digital state of the system

Consequently, the physical representation performs a dual function: On the one hand, it represents digital information and system states and, on the other, it serves as an interactive control element. Hence, the position, orientation or arrangement of a phicon, for example, can encode information about the corresponding digital state. At the same time, moving, rotating or connecting the object modifies that state. Thus, *representation* and *control* are connected not only conceptually but also physically [20].

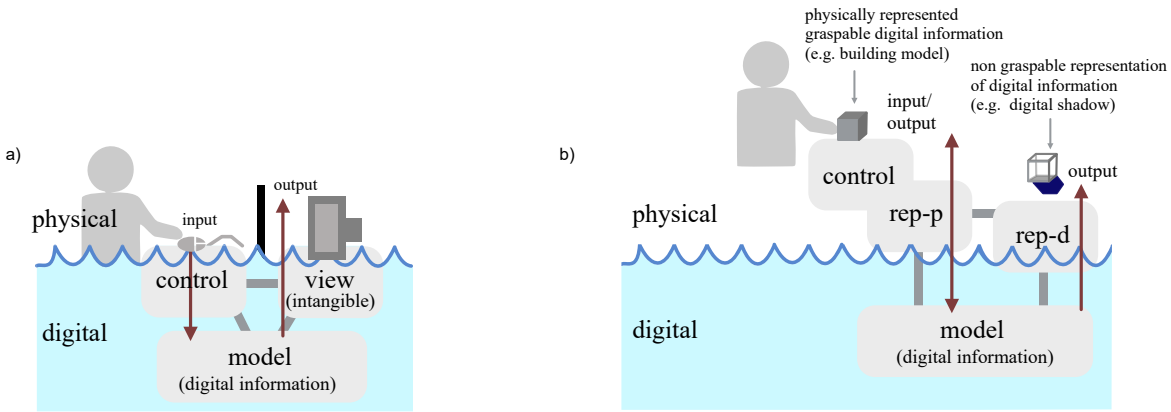

Figure 2: Interaction models for (a) a graphical user interface using the MVC model and (b) a tangible user interface using the MCRpd model [20]

2.3.2 Conclusion

The distinctive characteristic of a **TUI** lies in the close coupling between representation and action. Whereas a **GUI** typically separates the visual representation of digital information on a screen from interaction through an input device, a **TUI** uses physical artefacts both to represent digital content and to manipulate it [20]. This coupling can support the constructivist and constructionist learning

approaches discussed in Section 2.1 by enabling learners to explore relationships through physical manipulation and to externalise their ideas in the form of tangible, directly modifiable artefacts [10].

2.4 Design Factors for TUI

The comprehensibility and usability of a **TUI** depend fundamentally on how coherently its physical objects, their digital meanings and the actions available through them are coordinated.

2.4.1 Coupling Objects with Digital Information

Ullmer and Ishii provide an overview of various types of represented digital information, coupling methods and approaches to physical representation in order to describe the coupling between phicons and digital information [20].

Types of Represented Digital Information

A phicon in a **TUI** can represent different kinds of digital information. These include *static media*, such as images and three-dimensional models, *dynamic content*, such as video streams or computationally generated graphics, and *digital attributes*, including colours or material properties. Within a TUI, a physical object does not necessarily correspond to a single digital file but can have different digital meanings. For example, a physical figure placed on an audio player may represent an entire audiobook, while another object may control playback. To the user, these objects represent a story or an action rather than the underlying files and computational processes. What matters most is therefore the meaning of an object and the actions associated with it [20].

Methods of Coupling

In addition to the different types of information represented, coupling methods serve to categorise how and when the relationship between a phicon and its digital representation is formed:

- In *static coupling*, the meaning of an object is defined in advance. The physical object is permanently associated with a specific digital geometry, property or function and this association cannot be altered during interaction.
- By contrast, a *dynamic coupling* is established or modified during use. For instance, inserting a phicon into a specific slot or performing an action within a graphical user interface can associate it with new digital content or alter its existing assignment.
- *Interactive coupling* occurs when the meaning of an object changes depending on the other objects with which it is connected or combined.

Static coupling therefore provides a stable and often more easily understandable meaning, whereas *dynamic* and *interactive coupling* enable the same physical objects to be used more flexibly. Regardless of which coupling method is used, it is essential that changes to the physical object are reflected immediately and transparently in the digital representation. If this correspondence is delayed, missing or inconsistent, the relationship between the physical and digital representations becomes difficult to understand [20].

Approaches to Physical Representation

Both the temporal precision of the coupling and the appearance of the physical object influence how clearly its digital meaning and possible interactions can be understood. Four approaches to physical representation are presented in Table 3. These approaches are not necessarily mutually exclusive and may be combined within a single system [20].

Approach	Description	Illustrative Example
Found objects	Existing objects augmented with technology and placed in a new digital context	A sensor-equipped toy car moved across a table to control a digital vehicle

Approach	Description	Illustrative Example
Engineering-driven	Physical form determined by electronic and mechanical requirements	Appearance of a prototype determined by its electronic components
Design-driven	Physical form designed to communicate meaning and indicate possible interactions	Tokens shaped like musical instruments to select the corresponding digital sounds
Work-practice-based	Design based on existing physical artefacts from specific professional or application contexts	Digitally tagged library cards retain the familiar process of assigning borrowed books to individual users

Table 3: *Approaches to physical representation in a TUI [20]*

Conclusion

The coupling between physical objects and digital information can differ in content, flexibility and physical representation. Selecting an appropriate approach therefore requires a careful consideration of the intended users and their abilities, as well as the specific application context. Regardless of the chosen approach, an effective TUI requires a clear and immediately perceptible relationship between the phicon and the associated digital meaning. The design of this coupling therefore directly influences the system's comprehensibility, usability and flexibility.

2.4.2 The *Token and Constraint* Framework

Once digital information has been assigned to a phicon, it must be determined how this object can be physically arranged and how changes in its configuration can be translated into digital operations [19].

One approach to structuring tangible interaction is the *Token and Constraint* framework. The discrete physical objects that represent digital information are referred to as *tokens* within this framework. They correspond to the physical representations previously introduced as *phicons*. However, the term *token* is retained in this section to remain consistent with the framework's original terminology. *Tokens* are placed or manipulated within physical structures known as constraints. These constraints define where and how a token can be positioned or moved and are mapped to specific digital operations [19].

Phases of Interaction

The interaction within *Token and Constraint* systems can involve two phases: *Associate* and *Manipulate*. During the *Associate* phase, a *token* is placed within a constraint, thereby establishing a logical relationship between the information represented by the *token* and the operation assigned to the constraint. This initially creates a discrete state, such as presence or absence, activation or deactivation, or selection and deselection. The same *token* may be inserted into different constraints in order to associate its digital information with different predefined operations [19].

The *Manipulate* phase starts once the *token* has been physically engaged with the *constraint*. The *token* can now be moved within the *constraint*'s mechanical boundaries to modify parameters of the associated operation. The absolute and relative positions of *tokens*, both in relation to the *constraint* and to one another, can result in different digital interpretations. The meaning of an interaction therefore emerges from the interplay between the *token*, the *constraint* and the overall physical configuration. Translating or rotating the *token* may result, for example, in the continuous adjustment of a value, the navigation through a list, or the alteration of another variable of the digital system. However, not every *Token and Constraint* system needs to support both phases of interaction, since in some systems, inserting and removing a *token* may already constitute the complete interaction [19].

Physical Syntax and Interaction Guidance

Physical constraints influence not only the digital interpretation of an action but also its haptic and perceptual qualities, since the physical syntax renders some of the available interaction possibilities directly perceivable. In this context, syntax refers to the rules according to which individual phicons can be combined to form an input that can be meaningfully interpreted by the system. The geometry and mechanical configuration of the constraints function as a physical grammar by directly indicating which actions are possible and which are prevented. This supports the creation of physical configurations that are interpretable both by the user and by the underlying computational system. By reducing the range of possible manipulations, invalid states can be avoided and the available courses of action can be made easier to understand [19].

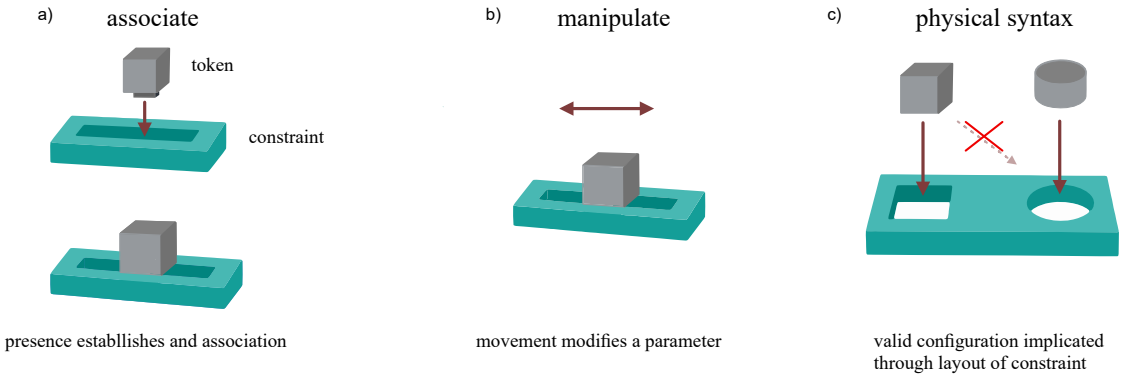

Figure 3: Interaction phases and physical syntax within the Token and Constraint framework

Styles of Mapping

Mapping styles describe how the resulting physical configuration is translated into a computer-interpretable representation. Three principal styles can be distinguished: *spatial*, *relational* and *constructive* mapping.

In the *spatial mapping*, interpretation is based on the geometric position, distance, or orientation of phicons in the physical space. The pivotal factor, therefore, is the object's position within a spatial frame of reference.

In the context of *relational mapping*, the absolute position of a phicon is less significant than its relationship to other phicons or to a particular *constraint*. The digital interpretation may therefore depend on properties such as sequence, proximity, adjacency, or assignment.

In *constructive mapping*, the digital meaning of a given structure is derived from the manner in which several physical components are assembled into a larger structure. The interpretation is therefore not based on a single element, but rather on the resulting overall physical structure.

These mapping styles are not mutually exclusive and may be combined within a single **TUI**. An assembled structure may, for example, be interpreted based on both the connections between its components and their spatial positions or orientations [19].

Conclusion

The *Token and Constraint* framework demonstrates how physical actions can be translated into comprehensible digital operations. *Tokens* enable direct manipulation, while constraints guide interaction and prevent invalid configurations. These characteristics can support constructivist and constructionist learning by enabling structured exploration and making the consequences of physical actions observable. However, their educational value ultimately depends on the design of the learning task.

2.5 Feedback for TUI

In a **TUI**, feedback informs users whether an action has been detected, how it has been interpreted and how it has affected the state of the system. It thereby closes the perception-action loop between physical manipulation and digital processing and supports the independent use of the system [21, 22].

2.5.1 Feedback Modalities

As shown in Table 4, a system can communicate the consequences of an action through different output modalities. The physical properties of a **TUI** already provide intrinsic feedback that arises directly from manipulating its objects. Additional system-generated feedback may be presented visually, auditorily, haptically or through a coordinated combination of these modalities [22].

This multimodal feedback does not imply presenting as many signals as possible simultaneously, since excessive or unnecessarily complex information may increase processing demands and impede the interpretation of the feedback [23]. Instead, different modalities should offer alternative ways of accessing the same or complementary information, which can then be combined when a perceptual channel cannot be used or can only be used to a limited extent [24, 25].

Regardless of its modality, feedback can also be characterised along several dimensions concerning its location, temporal relationship to the action, dynamics and duration [22].

- Local or global: presented directly at the manipulated object or through a separate display
- Discrete or continuous: confirming an individual event or continuously indicating a state
- Immediate or delayed: responding directly to an action or providing a later summary
- Persistent or transient: remaining available until a correction is made or disappearing after a short period

Feedback Form	Function	Examples
Intrinsic	Provides information arising directly from the physical action	Mechanical resistance, form-fit connections or snapping into place
Visual	Communicates states, errors and changes visually	LEDs, displays, projections, or animations
Auditory	Attracts attention and communicates information auditorily	Sounds, signals, or spoken information
Haptic	Makes system responses physically perceptible	Vibration, movement, or force feedback
Multimodal	Provides complementary or alternative access through different perceptual channels	Coordinated combinations of visual, auditory, or haptic signals

Table 4: *Feedback modalities in tangible user interfaces*

The distinction presented in Table 4 is based on the differentiation between inherent and additionally generated information proposed by Wensveen et al. [22].

2.5.2 Feedback in Learning-Oriented TUIs

In learning-oriented TUIs, two levels of feedback should be distinguished:

- *Interaction feedback* communicates whether and how a physical action has been technically detected and processed [21].
- *Pedagogical feedback* evaluates the action or its outcome in relation to the learning task [23, 26].

A system may therefore detect an action correctly while evaluating it as incorrect in relation to the task. Conversely, a failure to detect an action must not be presented as an error made by the learner. Reliable interaction feedback is therefore a prerequisite for meaningful pedagogical feedback [21, 24].

Interaction Feedback

Interaction feedback makes the technical response of the system comprehensible. It should indicate whether a physical action has been detected and interpreted correctly. If an action cannot be processed, the system should distinguish a technical recognition problem from an incorrect solution to the learning task. Otherwise, learners may incorrectly attribute system failures to their own actions and become uncertain about how to proceed [21, 24].

Pedagogical Feedback

While *interaction feedback* primarily confirms whether an action has been technically processed, *pedagogical feedback* concerns the meaning of that action within the learning process. It supports learners in assessing their current progress, recognising deviations from the intended outcome and adapting their subsequent actions [26, 27].

Pedagogical feedback can fulfil different functions:

- Outcome feedback indicates whether a result is correct or incorrect.
- Process feedback provides information about the approach taken or a possible cause of error.
- Feedforward offers guidance for the next step.
- Progress feedback indicates what has already been achieved in relation to the intended goal.

Additionally, as it would restrict independent planning and problem-solving, the system should not prescribe every step, but provide graduated feedback in which support increases according to the learner's needs. This preserves opportunities for independent error analysis while preventing learners from becoming unable to continue because the feedback is insufficient or incomprehensible [23, 26].

2.5.3 Conclusion

Effective feedback in a learning-oriented **TUI** must make both the technical processing and the pedagogical meaning of an action comprehensible. By combining accessible feedback modalities with graduated support, the system can help learners recognise deviations and adapt their actions to work independently without prescribing the complete solution.

2.6 Child-Appropriate TUIs in the Field of Special Education

Whether a **TUI** is perceived as accessible, exploratory and supportive of learning depends not only on its general design and feedback, but also on its adaptation to the specific needs of children. A literature review of TUIs for children's creative learning highlights the factors of abstractness, openness, richness and complexity, as well as a smooth experience. Since these factors were also not specifically formulated for students with a special educational focus, they are discussed below with consideration of the heterogeneous learning abilities and conditions for action within the defined target group [15].

Abstractness

The factor of *Abstractness* describes how concrete or ambiguous the meaning of a phicon is designed to be. While abstract and not fully predetermined representations can promote imagination and divergent thinking, an overly abstract design can lead to confusion, particularly among children without the required prior knowledge or additional explanation. The choice of physical representation and the manner in which it is coupled with digital information, explained in Section 2.4.1,

therefore directly influence how comprehensible users perceive the meaning of the phicon to be [15].

For students within the selected special educational focus, a balance between abstraction and clarity is therefore required. Phicons should provide sufficient indications of their function and possible use, but at the same time, the design should not be unnecessarily simplified or infantilising, so that it still provides sufficient depth for exploration [1–3].

Openness

Openness defines the scope that a TUI provides for individual actions, solution strategies and interpretations and is indirectly influenced by the number and restrictiveness of its physical *constraints* (Section 2.4.2). While excessive instruction can restrict creativity, insufficient support or scaffolding often leads to frustration [15].

For students with a special educational focus on **GE**, **LE** and **ASD**, *structured openness* is particularly relevant. Clear intermediate goals can prevent children from becoming overwhelmed. When these are supplemented by step-by-step support that is adapted to the individual learner's current level and can be reduced as confidence increases, a comprehensible framework for action is created without providing excessive instruction [1–3].

Richness and Complexity

While *Richness* refers to the variety of possible content, actions and combinations within a system, *Complexity* describes the demands that arise from this variety. A design that is too simple does not provide sufficient inspiration, whereas overly complex interfaces can overwhelm children. The system should support children in overcoming difficulties independently without removing the challenge entirely [15].

Due to the different learning conditions associated with the considered support needs, the complexity of the **TUI** should not be reduced uniformly but should instead be individually adaptable. Different levels of difficulty can be designed to introduce new content and possibilities for action gradually. The objective should therefore not be to create the simplest possible design, but to provide an appropriate level of challenge for each child [1–3].

Smooth Experience

Faulty hardware or software can severely impair the learning experience. Unreliable systems make children feel uncertain and discourage them from learning from mistakes or taking risks, particularly when it is not clearly recognisable whether an error was caused by the system or by their own actions [15].

For students with special educational needs, this can considerably reduce both motivation and confidence in acting independently. The feedback described in Section 2.5 should therefore be reliable, immediate and clearly attributable. Multimodal feedback can provide access through different perceptual channels, but its intensity and quantity must remain limited or adaptable in order to prevent unnecessary sensory overload [1–3].

The factors discussed above originate from the literature review on **TUI** for children's creative learning. Based on the learning needs of students within the considered areas of special educational support, two additional factors are proposed: *Adaptability* and *Multimodal Accessibility*.

Adaptability

Adaptability describes the extent to which a TUI can be adjusted to learners' individual abilities and needs. Task difficulty, pace and level of support should be configurable without changing the fundamental interaction concept. Such adaptations can provide sufficient guidance while allowing learners to work as independently as possible [1–3].

Multimodal Accessibility

Multimodal Accessibility requires relevant information and feedback to be available through more than one perceptual channel. Information should therefore not be communicated exclusively through text, colour, or sound, as each of these may create barriers for individual learners. Combining these modes, however, can provide alternative means of access. The number and intensity of stimuli should remain adaptable to avoid sensory overload [4].

Teachers

For use in schools, the requirements of teachers must also be considered and teachers should be included as stakeholders in the TUI design process. Nevertheless, the system should be usable without programming knowledge or time-consuming calibration. Tasks must be easy to select and understand, the components must be transportable and the system must be capable of being set up with minimal preparation time [15].

Conclusion

The design factors show that there is no universally child-appropriate TUI. Its suitability depends on how well the interaction design responds to individual abilities, support needs and sensory preferences. The central design challenge is therefore to provide sufficient structure and accessibility without unnecessarily restricting independent exploration.

3 Related Work

Physical building brick systems have become a promising form of **TUI** because their material properties and haptic feedback allow users to manipulate digital information through familiar physical actions. Existing approaches can broadly be divided into systems employing passive bricks and systems that use active bricks [28, 29]. In this context, passive bricks do not perform local computation. Instead, their identity, position and connectivity within a construction are determined by an external sensing system [30]. Active bricks, by contrast, contain embedded electronic components and actively participate in the detection process [31, 32].

The following sections review concepts based on these two approaches to establish the relevant state of the art.

3.1 Passive Brick Systems

Passive bricks can be detected using different sensing principles, including computer vision, capacitive sensing and Radio Frequency Identification (**RFID**).

Computer-vision-based systems use one or more cameras to determine the structure constructed from the physical bricks. *DuploTrack* employs such an approach: By analysing the depth data and colour information of the bricks using an RGB-D camera, it can provide real-time assistance during the construction process of *LEGO DUPLO* models. A general limitation of such camera-based detection is that the relevant parts of the construction must remain visible. Bricks that are temporarily occluded by other bricks or the user's hands therefore result in incomplete or incorrect detection. Moreover, the camera usually has to be positioned and calibrated according to the dimensions and lighting conditions of the interaction area [33].

By contrast, an infrared camera is used by Baudisch et al. (2010). For the *Lumino* bricks, the system relies on visual markers and a light-conducting fibre-optic bundle that transfers the fiducial marker to the surface of an interactive table. The table's internal infrared camera then captures the resulting composite marker image and uses it to reconstruct the stack. However, due to the high light loss at the junctions between the optical fibres, the stacking height is limited to five layers [30].

CapStones uses capacitive sensing to extend applications on conventional touchscreens with graspable physical objects. The Three-Dimensional (**3D**) arrangement of the game pieces can be detected on conventional devices by "channelling" the user's capacitance through a system of internal wires and connecting pins from the top of a stack to the touchscreen. Users can place the blocks on the screen, move them around and combine them into vertical stacks. However, as the capacitance of the internal wires exceeds the sensor's threshold when they are extended, the maximum stack height is limited to three to four layers. Furthermore, the blocks must always be touched to be visible, as current tablet sensors are calibrated to specific touch detection thresholds [34].

In the *ProtoBricks* system, capacitive detection is supplemented by a colour sensor. While capacitive touch sensors integrated into the baseplate detect the user's body through the conductive blocks and determine their spatial placement, the colour sensor detects the colour and thus the type of the brick that has been placed. However, it is not possible to capture information regarding the precise rotation of the individual bricks. Furthermore, the bricks must be placed one at a time; otherwise, the system will recognise only a single action [35].

Besides camera-based and capacitive sensing, physical brick constructions can also be detected using **RFID**. In such systems, the bricks are considered passive because they operate without batteries and obtain the required energy wirelessly from an **RFID** reader.

The *RFIBricks* system uses such a radio-based recognition principle with modified Ultra High Frequency (**UHF**) **RFID** tags. Each tag is separated into an antenna and a chip component, which

are connected through conductive neodymium magnets. The resulting circuit functions as a contact switch: the tag becomes readable only when the corresponding physical connection between two blocks is established. The relative orientation of two building blocks is detected by the four **RFID** contact switches located on the surfaces of the cubes, each of which triggers a specific ID combination depending on the rotation. However, to avoid excessive complexity and tag density, these contacts are located only on the top and bottom surfaces. The resulting constructions are therefore limited to one-dimensional stacks [28].

RFIMatch is closely related to *RFIBricks* and similarly combines modified UHF RFID tags with magnetic elements. In contrast to *RFIBricks*, *RFIMatch* is designed to detect associations without requiring precisely aligned physical contacts. Its sensing mechanism is therefore less dependent on the exact orientation and alignment of the interacting objects [36].

Since both approaches operate in the **UHF** frequency range, the required readers can be comparatively expensive, large and power-intensive [29]. To overcome this issue, the *NFCStack* concept uses a lower-cost MFRC522 Near Field Communication (**NFC**) reader. A multiplexed reader in the base station transmits its signal through antenna extensions integrated into the bricks. Depending on the brick design, this signal is relayed through transmission lines or inductively coupled coils towards the top of the stack. As with *RFIBricks*, it is possible to identify the individual blocks and determine their position and orientation within a stack. However, its physical connectivity is also restricted to vertical stacking and therefore supports only one-dimensional constructions [29].

All three **RFID** concepts share two limitations: The limited transmission bandwidth of the technology means that, in all three concepts, the system becomes less reliable when there are too many active tags operating simultaneously. Furthermore, the achievable stack height is limited because the available signal strength decreases with increasing distance from the reader and with every additional coupling interface [28, 29, 36].

3.2 Active Brick Systems

Unlike passive systems, active bricks integrate electronic processing and communication components into the individual bricks. The bricks can therefore contribute directly to detecting the construction. To achieve this, the resulting topology information can be either transferred through the physical chain of bricks or transmitted wirelessly by the individual bricks.

E-Block, introduced by Wang et al. (2013), follows such a principle. *E-Block* is a tangible programming tool in which children arrange physical bricks into a sequence to specify the path of a virtual character. When activated, each brick transmits its identifier wirelessly to a receiver connected to the host computer. Communication between neighbouring blocks is limited to a sequential activation process that follows a parent–child logic: A parent brick uses an infrared signal to activate an adjacent child brick. The child then transmits its identifier to the computer and enables its own infrared transmitter to activate the next brick in the sequence. Because the connected bricks transmit their identifiers one after another, the host computer can infer their order from the arrival times of the messages and reconstruct the one-dimensional structure. This limits the representable topology to a sequential arrangement [37].

The *i-Cube*, a digital educational tool designed to support cognitive development, uses a related sequential round-robin principle to identify neighbouring relationships. Neighbouring bricks are detected locally using inductive sensors and a fire-and-latch protocol. In this protocol, the transmitter on each face of every brick in the system is sequentially activated to capture the complete structure. To detect the relative orientation, each face contains four symmetrically arranged inductive receivers instead of one. When a transmitter on an opposing face of a neighbouring brick is activated, one of these receivers is triggered. This enables the system to identify the connected faces and the rotation angle between the two bricks. This approach allows three-dimensional construc-

tions to be detected, including the local orientation of individual bricks. Four magnets arranged on each brick face ensure that the contacting faces remain physically attached [38].

The concept of *roBlocks* also uses magnets, employing them as both mechanical connectors and electrical contacts. Once connected, the bricks form an ad hoc communication network through which they exchange and broadcast data. A dedicated communication brick called *Comm-Block* collects network information and forwards it wirelessly to the host computer. This special brick also initiates topology discovery by querying the connected bricks and collecting reports on their local connections. In addition to the Base brick, a Power brick supplies electrical energy to the network via the magnetic contacts. Consequently, the individual bricks do not require separate batteries. This distinguishes the system from previously discussed active-brick approaches, in which every brick must be charged or equipped with a replaceable battery [31].

A comparable central power-supply principle is also used in *ActiveCube*, where the Base brick provides the required energy. The electrical connection between adjacent bricks is established through three concentric terminals located at the centre of each brick face, which are used for the communication network and ground, as well as additional mechanical connection hooks that provide the power supply. The ActiveCubes form a wired RS-485 network and communicate using LON technology. When a new brick is attached, both participating cubes report the connection, including their brick identifiers and connected brick faces, to the brick base. The bricks obtain information about the connected face through one of the connection hooks used for the power supply. The child detects the voltage applied to this input hook and can immediately determine which of its six faces is connected. The host computer combines the different Identification (**ID**) messages to update the connection-status tree that represents the topology of the physical three-dimensional construction [32].

Figure 4 illustrates the brick-face interfaces of *i-Cube*, *roBlocks* and *ActiveCube*, which represent different approaches to mechanical connection, electrical contacting and local connection detection.

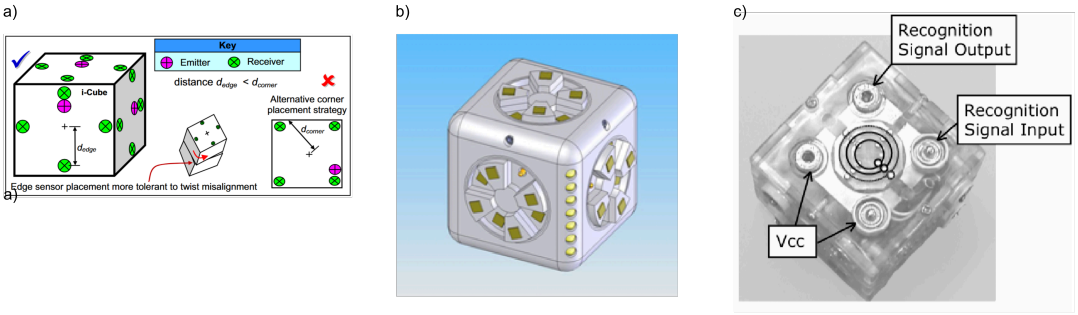

Figure 4: Brick-face interfaces of (a) *i-Cube*, (b) *roBlocks* and (c) *ActiveCube* [31, 32, 38]

3.2.1 Integrated Sensors and Actuators

In addition to contributing to topology detection, active bricks can also support local output and additional sensing: *ActiveCube* includes functional cubes containing components such as gyroscopic sensors, vibration actuators and Light-Emitting Diode (**LED**) matrices [32]. *i-Cube* provides illuminated faces and local audio output through vibration speakers [38]. *roBlocks* includes specialised actuator blocks equipped with geared motors, piezoelectric speakers or LEDs, whereas *E-Block* uses an LED on the upper surface of each brick to communicate its current status [31, 37].

3.3 Comparison of Active and Passive Brick Systems

Active and passive brick systems present different advantages and limitations. Passive blocks operate without individual batteries and do not perform local computation. This reduces maintenance at the brick level and makes them particularly suitable for systems involving large numbers of physical components. Their comparatively simple internal construction may also reduce the cost

of producing individual blocks. However, passive blocks depend on external sensing infrastructure. Camera-based systems like *DuploTrack* are susceptible to occlusion when blocks are covered by other objects or by the users' hands. They may also require calibration and specialised camera or display hardware. Capacitive and optical approaches, as well as approaches based on **RFID**, can be limited by signal attenuation, reading range or the maximum detectable stack height.

Active blocks contain embedded electronic components, such as microcontrollers, communication modules and, depending on the system, individual batteries. Because they participate directly in the recognition process, they generally enable reliable detection of stacking operations and connections between neighbouring blocks. Depending on the system, this can include the detection of three-dimensional constructions and the local orientation of individual bricks, as demonstrated by *i-Cube*. Their local electronics and power supply can additionally support integrated sensors, actuators and displays, which can provide rich multimodal interactions. However, these capabilities increase both the complexity and the cost of each brick. The required processors, sensors and communication modules increase manufacturing costs and can limit scalability when large numbers of blocks are needed. In systems such as *E-Block* that depend on physical cables or wired connections, the users' freedom of movement may be restricted. Battery-powered systems like *E-Block* and *i-Cube* require regular charging or battery replacement, resulting in additional maintenance effort. Watanabe et al.'s *ActiveCube* (2004) and Schweikardt and Gross's *roBlocks* (2006) reduce battery maintenance by distributing power through electrical contacts between connected blocks, although this still requires more complex brick hardware.

The choice between passive and active blocks therefore represents a trade-off between brick-level simplicity and the recognition and interaction capabilities of the overall system.

4 Requirements Analysis

The requirements analysis translates the findings from the theoretical background and the design principles derived from the special educational context into concrete requirements for the prototype. Following a constructionist approach, the objective is to develop a modular, brick-based **TUI** in the form of a learning game suitable for use in special education. The prototype is intended to enable students to complete three-dimensional construction tasks independently while receiving immediate feedback on whether their construction corresponds to the target structure or not.

The requirements are derived from three perspectives:

- The learning prerequisites of the target group and the resulting educational implications
- The characteristics and design principles of tangible user interfaces discussed in the theoretical background
- The technical, financial and time constraints of prototype development

The following sections describe the context of use, the application scenario and the core requirements guiding the system's design and development.

4.1 Context of Use and Didactic Application Scenario

The primary target group consists of students with special educational needs related to **GE**, **LE** and **ASD**. Teachers and other educational professionals represent a further key stakeholder group, as they integrate the learning game into classroom activities and support the learning process.

By enabling students to complete three-dimensional construction tasks independently, the learning game is intended to address the following competencies:

- Primary: spatial thinking, action planning and self-directed learning
- Secondary: problem-solving, error detection and self-correction
- Additional: fine motor skills, concentration and hand–eye coordination

The primary application scenario is a spatial reconstruction task, where students select a three-dimensional target structure appropriate to their individual level and attempt to recreate it using the available bricks. Although this type of task can also be carried out using conventional construction materials, the proposed system extends the building process through automated feedback and can indicate whether the construction is correct or needs to be modified.

The level of difficulty can be varied through the number of bricks, their spatial arrangement and the required rotations. Simple tasks may consist of a small number of bricks connected primarily in a linear arrangement, whereas more challenging tasks may involve multiple spatial directions, different orientations and branching structures. Future task types could include completing unfinished structures, identifying incorrect connections, or constructing freely without a predefined target.

4.2 Pedagogical and Interaction-related Requirements

In accordance with the constructionist understanding of learning described in Chapter Section 2.1.3, as well as the *embodied cognition* theory of Section 2.2.1, the active creation of a physical artefact must form the core of the learning activity [TASK1] . Learners must be able to observe, inspect and directly modify the emerging construction throughout the task [IA1] . This enables them to translate their mental representation into a concrete spatial structure through which the effects of individual actions become observable. To support independent use, the system must also be operable without technical or programming knowledge and without requiring recalibration [IA2] .

At the same time, the tasks must accommodate the diverse learning prerequisites of the target group explained in Section 1.1 and thus provide a clear and manageable framework for action [IA3] , without describing every individual action. Processes and sub-goals must be presented in a way

that is easy to follow, whilst allowing sufficient time for completion [IA4] and opportunities for repetition [IA5] .

The design factors described in Section 2.6 must also be considered. The principle of *structured openness* requires tasks to provide a clear framework while allowing learners to explore different approaches and solution attempts [TUI1] . Tasks must be structured in such a way that the objective remains clear and step-by-step guidance can be accessed where necessary. The factors of *richness*, *complexity* and *adaptability* further require the learning game to accommodate different levels of ability. This can be achieved by adapting the level of difficulty and scope of the feedback to different learning requirements as far as possible, so that the same basic principle of interaction can be applied to tasks of varying difficulty levels [TUI2] . Finally, the bricks must provide sufficiently clear physical cues for learners to understand their meaning and possible uses. Their physical design and digital functions must therefore be consistently aligned [TUI3] .

ID	Requirement
TASK1	The task must be completed through the active construction and modification of a physical artefact
IA1	The resulting construction must remain visible, inspectable and modifiable throughout the task
IA2	The system must be usable without technical or programming knowledge and without requiring recalibration
IA3	The objective and structure of the task must be presented clearly and comprehensibly
IA4	Learners must be able to complete the task at their own pace
IA5	Previously completed action steps must be correctable and repeatable
TUI1	The task should provide a structured framework while allowing different approaches and solution attempts within that framework
TUI2	The level of difficulty should be adaptable by varying the number of bricks, their spatial arrangement and the required rotations
TUI3	The physical design of the bricks and their digital functions must be consistently aligned

Table 5: *Educational and interaction-related requirements*

4.3 Requirements for Feedback and Assistance

To complete a construction task independently, learners must be able to determine whether their actions have been detected by the system and how those actions are evaluated within the learning task. In accordance with the distinction introduced in Section 2.5.2, the system must therefore provide both interaction feedback and pedagogical feedback [FB1] . Each feedback signal must be clearly attributable to the action that triggered it or to the affected part of the construction, allowing learners to understand what it refers to [FB2] . In this context, feedback and assistance should be available in graduated levels according to the learner's individual support needs [FB3] , [FB4] . As discussed in relation to multimodal accessibility in Section 2.6, providing feedback through multiple output modalities can improve accessibility and is therefore desirable [IA6] . However, when multiple modalities are used, their signals must be coordinated and consistent in order to avoid distraction.

ID	Requirement
FB1	Interaction and educational feedback must be ensured
FB2	Feedback must be clearly linked to the action that triggered it or the area concerned
FB3	Feedback should be graduated, increasing only when less support is insufficient for error correction
FB4	Assistance should be provided on a tiered basis, depending on the learner's level
IA6	Feedback should be provided through multiple output modalities

Table 6: *Requirements for feedback and assistance*

4.4 Geometrical and Ergonomic Requirements

The building blocks form the central element of the learning game: They serve as physical representations of digital information while simultaneously functioning as controls for interacting with the system (phicons). Their geometric and mechanical design therefore influences both the range of possible constructions and the comprehensibility and accessibility of the interaction. As already established by the educational requirements, the shape and connection mechanisms must clearly indicate how and where the blocks can be connected [TUI3] . To enable three-dimensional constructions, all faces of a block must provide connection capabilities [C1] . These connections must also allow two connected bricks to be positioned in different orientations relative to one another [C2] . Building on this structural flexibility, further variety could be introduced through additional block types with different geometric shapes [G1] . In order to hold the structure together, the connection between the building blocks must be stable enough to be undone with minimal effort, while also being strong enough to be created without requiring precise positioning that demands fine motor skills [C4] . In addition, each block must be capable of being securely grasped, positioned and connected using one hand [C3] .

ID	Requirement
G1	Different brick types could be implemented in different geometric shapes
C1	The bricks must be able to be arranged or stacked in three dimensions
C2	The bricks must be able to be positioned relative to one another in different orientations
C3	The bricks must be graspable and connected with one hand
C4	brick connections must be strong enough to hold the structure together and easy enough to disconnect with minimal effort

Table 7: *Physical design requirements*

4.5 Functional Requirements for Detection

The functional requirements define which information the system must detect and process in order to reconstruct the physical three-dimensional structure digitally: Each building block must be uniquely identifiable to enable reliable construction detection and targeted feedback [DET1] . In addition, the system must also determine which blocks are connected, through which faces they are connected [DET2] and how they are oriented relative to one another [DET3] . Only by combining these relationships can the system reconstruct the complete spatial configuration. To provide feedback on the constructed configuration, the system must compare the detected construction with the target structure [DET4] . The detection functionality should remain reliable when multiple blocks are connected simultaneously within a larger construction [DET5] . To prevent the system

from providing misleading information to users, ambiguous or contradictory configurations should be recognised as error states [DET6] .

ID	Requirement
DET1	The system must be able to uniquely detect each brick
DET2	The system must be able to detect which bricks are connected with each other and on what side the connection occurs
DET3	The system must be able to detect the relative rotation of two connected bricks in relation to each other
DET4	The system must be able to compare the detected brick structure with a target structure
DET5	The system should continue to operate reliably even when multiple bricks are connected simultaneously
DET6	Configurations that cannot be clearly identified or are contradictory should be reported as technical faults
DET7	Connections and changes to the structure must be detected reliably within a maximum of 400 ms

Table 8: *Requirements for construction detection*

The maximum response time of 400 ms [39], which has been specified for detecting changes to the construction, is initially a target value [DET7] . It is intended to ensure that an action and the resulting feedback are perceived as belonging together.

4.6 Electrical and Technical Requirements

The technical implementation should be as simple, modular and low-maintenance as possible, with individual components remaining replaceable at reasonable effort. To reduce maintenance requirements and avoid the need to charge multiple devices separately and thereby make the system suitable for use in schools, the building blocks must operate without individual batteries or rechargeable power sources [EL1] . For the initial proof of concept, the prototype should include at least ten building blocks, as this number is sufficient to demonstrate multi-part and spatially branching constructions [EL2] . In the future, the system should be expandable with additional blocks without requiring changes to its fundamental operating principle [EL3] .

ID	Requirement
EL1	The bricks must be operated without individual batteries or rechargeable cells
EL2	The system must consist of at least 10 bricks
EL3	The system should be extendable without changing any basic operating principles

Table 9: *Electrical and technical requirements*

4.7 Safety and Economic Requirements

The prototype is not a certified toy and may therefore only be used under supervision. Nevertheless, potential risks must already be reduced through the physical design and electronic components. This includes protection against overvoltage and short circuits [S4] , while potentially detachable small parts must be enclosed and protected against direct access [S2] . The building blocks must also withstand repeated grasping, connecting and disconnecting [S1] . In addition, each block should be easy to assemble and disassemble without damaging its components [S3] . The cost per building block should preferably remain between one and three euros. However, the total system cost cannot

yet be determined precisely, as it also depends on the base station, control electronics and potential output devices [EC1] .

ID	Requirement
S1	The bricks must be able to withstand repeated gripping, connecting and disconnecting
S2	Electronic components and small parts that could come loose must be protected from direct access by the brick housing
S3	The bricks should be assembled and disassembled easily without being damaged
S4	The design of the bricks must include protection against overvoltage and short circuits
EC1	The total cost of a brick should not exceed 3 euros

Table 10: *Safety and economic requirements*

5 Concept Development

The requirements analysis conducted in Section 4 provides the foundation for the system concept. The objective of the concept development is to translate the individual pedagogical, interaction-related, functional, mechanical and technical requirements into a coherent overall system.

5.1 Usage Scenario

The usage scenario describes the general sequence of a guided construction task from the learner's perspective: At the beginning, a target structure is selected from a predefined variety of tasks in the learning application and presented as a reference. It is selected either by the teacher or, where appropriate for the learning context, by the learner. Afterwards, the learner begins by connecting the first cube to the baseplate and then adds further cubes until the specified target structure has been reconstructed. Previously completed steps can still be modified, as cubes can be detached, repositioned and reconnected at any point during the task. Following each modification, the system detects the current physical configuration and updates its digital representation, which is then compared with the target and, depending on whether it is correct, results in one of two outcomes: If the current configuration constitutes a valid part of the target structure, the learner can continue with the next construction step. However, if a deviation is detected, the system provides corresponding feedback so that the affected step can be reviewed and corrected. The task is completed once the physical construction fully matches the target structure. It can then either be repeated or a new target structure is selected. The activity diagram in Figure 5 illustrates the sequence of a single construction task and therefore terminates once that task has been completed.

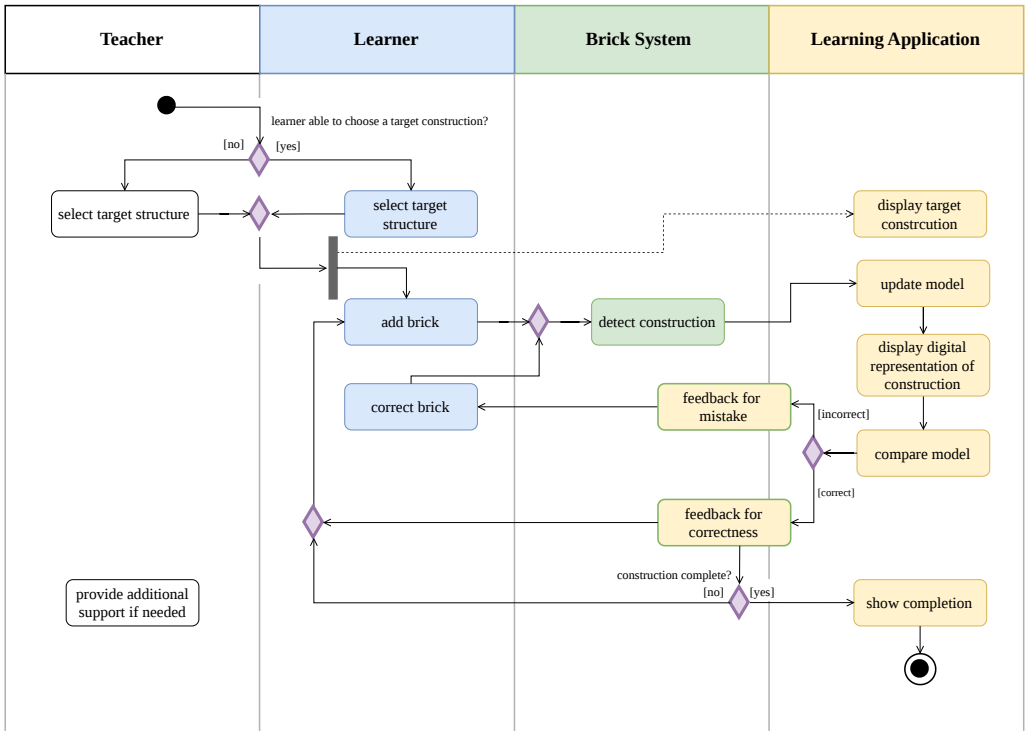

Figure 5: Activity diagram of a single guided construction task

The task is completed at the learner's individual pace and does not impose a fixed time limit. Additionally, a specific construction sequence is required only where it necessarily follows from the spatial dependencies of the target structure.

Throughout the process, the teacher retains responsibility for the pedagogical guidance of the learner and can adapt the available selection of tasks or provide additional explanations if the feedback or assistance offered by the system is insufficient or not understood. Automated feedback is therefore intended to support the teacher during periods of independent work and should not replace individual pedagogical guidance.

5.2 System Architecture

To support the usage scenario described above and address the associated detection and feedback requirements, the proposed architecture uses an active brick system in which the building bricks participate directly in detecting the physical construction and providing local feedback. The baseplate coordinates the physical brick system and mediates communication with the learning application. The learning application forms the digital part of the system and supports the presentation and evaluation of the construction task.

This high-level allocation of roles is translated into five functional subsystems, which structure the subsequent concept development:

- Physical Design and Connection Subsystem: Provision of the physical building bricks, housings and mechanical connections
- Electronic Connection and Power Subsystem: Identification of individual building bricks, detection of connections and relative rotations, and provision of the power supply
- Data, Communication and Software Subsystem: Transmission of the detected structural data, generation of a digital structure model and spatial reconstruction of the physical construction
- Task and Interaction Subsystem: Selection and presentation of the target structure and provision of the basic task workflow
- Visualisation and Feedback Subsystem: Comparison of the detected and target structures, visualisation of the current construction, and generation and output of feedback

The architecture means that a physical action cannot be interpreted directly by the learning application. Instead, the connection must first be established mechanically and detected electronically before its local connection data is transformed into a digital model of the overall structure. Only once this model is available can the learning application compare the detected construction with the selected target structure and generate task-related feedback.

The design decisions underlying this architecture are developed and justified in the following subsystem sections.

System Components and Responsibilities

The functions identified in the preceding functional decomposition are assigned to the components of the overall system. The technical system consists of:

- Several active bricks
- One baseplate
- One digital computer-based learning application

Learners and pedagogical professionals are external actors and therefore not part of the technical system components. Table 11 provides an overview of the elements of the overall system and their primary responsibilities, whereas Figure 6 illustrates the principal interactions between them.

Element	Classification	Primary Responsibility
Learner	External actor	Select the target structure, assemble the current physical structure, respond to feedback
Teacher	External actor	Adapts the selection of tasks, guides the learning process, provides additional support
Building bricks	Technical component	Form the physical construction, provide local connection data, present local feedback
Baseplate	Technical component	Provide the starting point and central power supply, mediate between the brick system and the learning application
Learning application	Technical component	Provide the target structure, process structural data, visualise the current construction, compare it with the target structure, generate feedback

Table 11: *System components and responsibilities*

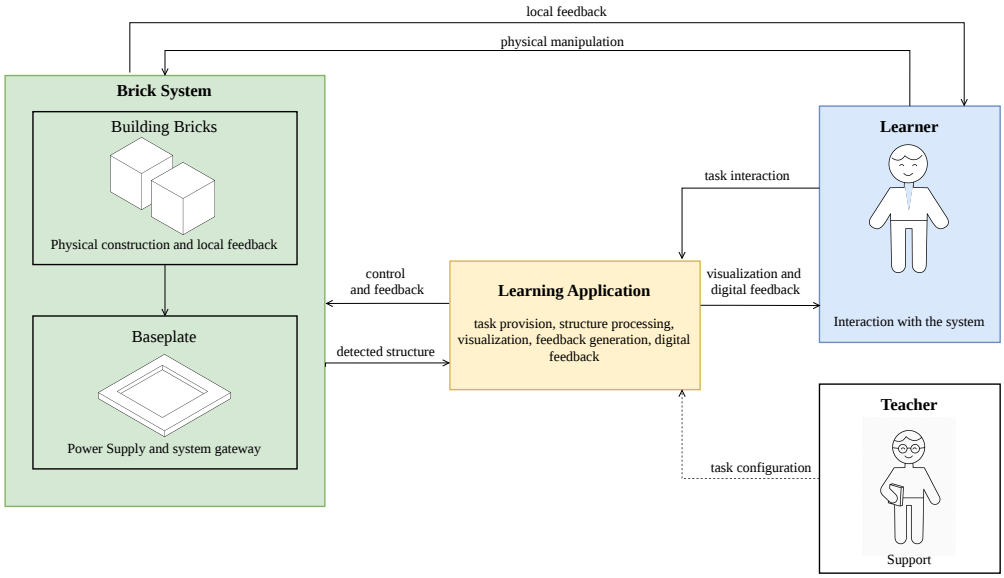

Figure 6: High-level architecture of the proposed brick-based learning system

5.3 Conceptual Definition of the Levels of Connection

Within the proposed TUI, each building brick functions as a phicon: it represents an element of the construction, while its physical manipulation simultaneously provides input to the digital system. To describe how these phicons are connected unambiguously, a distinction is made between the mechanical, electronic and logical connection levels. Each building brick has one docking face through which it is connected to the existing construction and, depending on its type, one or more receiving faces to which further building bricks can be attached. The docking face contains the electronic upstream interface, whereas each receiving face contains a downstream interface. In the digital model, the building brick already connected to the system is represented by the parent node, while the newly added building brick is represented by the child node. A connection can therefore be described at the three levels presented in the following Table 12:

Level	Term	Definition
Mechanics	Connection face	General term for all physical brick faces through which a connection can be established
Mechanics	Docking face and receiving face	Connection faces of the building bricks
Electronics	Upstream and downstream interface	Electrical contacts located on the connection faces
Data model	Parent and child nodes	Directed relationship within the digital tree
In general	Parent brick	Building brick that is already connected to the system
In general	Child brick	Building brick whose docking face is connected to a receiving face of the parent brick

Table 12: *Terminology across the connection levels*

5.4 Physical Design and Connection Subsystem

Physical interaction with the system begins with the building blocks and their connections. Their design shapes how learners experience the construction process while also determining how reliably the resulting physical arrangement can be interpreted by the technical system. This section therefore describes the physical design that forms the tangible foundation of the system. In accordance with the terminology defined in Section 5.3, the terms docking face and receiving face are used throughout the following sections.

5.4.1 Basic Geometry and Block Types

In accordance with requirement [C2] , the connection faces must allow two connected bricks to assume a discrete and unambiguously interpretable relative orientation. Existing systems such as *i-Cube*, *RFIBricks* and *NFCStack* similarly employ fourfold rotational symmetry to distinguish the orientation of connected bricks [28, 29, 38]. Based on this principle, a standardised square geometry is selected for all connection faces. This geometry permits a rotationally symmetrical contact arrangement and provides four mechanically valid orientations in increments of 90°. Based on the standardised connection geometry, two brick types are defined in accordance with [G1] :

- The cube brick with six square connection faces
- The smashed brick with two opposing square connection faces, while its remaining faces cannot be used to establish connections

Whereas the cube brick supports linear as well as branched two- and three-dimensional constructions along its three mutually orthogonal spatial axes, the smashed brick permits only linear connections along its local z-axis. Figure 7 illustrates the two brick types.

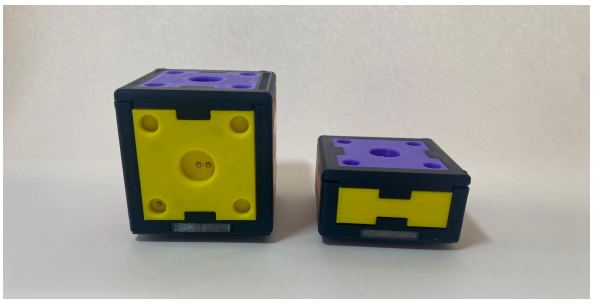

Figure 7: Comparison of the two brick types: cube brick (left) and smashed brick (right)

Directed Design of the Connection Faces

To limit the complexity of manufacturing and connection detection, two functionally distinct types of connection faces are defined. The docking face connects a brick to the existing construction, whereas the receiving faces allow additional bricks to be attached through their respective docking faces. To enable three-dimensional constructions as required by [C1] , the cube brick must provide connection possibilities along all three of its local spatial axes. As one of its six connection faces serves as the docking face, the remaining five are designed as receiving faces. Because the smashed brick has only two opposing connection faces, it can have only one receiving face. Consequently, a valid connection can only be established between a receiving face of the parent brick and the docking face of the child brick.

To identify the intended target connection and orientation between two bricks, the receiving faces are differentiated by colour. This directed design provides the physical basis for representing the resulting construction as a parent–child structure. The corresponding digital structure model is described in Section 5.6.2.

Mechanical Connection Principle

The mechanical connection principle was selected with regard to stability, unambiguous alignment and the low operating force specified by [C4] . For the prototype, a combination of a mechanical form-fit mechanism and magnetic assistance is used: The docking face contains recessed guiding elements, while the receiving faces contain corresponding protruding elements. Magnets with complementary polarities draw compatible faces together and stabilise the resulting connection without requiring the force associated with a conventional snap-fit mechanism.

Together, the guiding elements and magnets implement a mechanical constraint as described in Section 2.4.2. They align the electrical contact areas and impede invalid connections because incompatible interface geometries do not fit and equally polarised magnets repel one another. The available connection possibilities are therefore partly conveyed through the physical design itself.

5.4.2 Housing and Maintainability

The housing must completely enclose the electronic components and magnets and prevent small parts from becoming detached unintentionally, as specified by [S2] . At the same time, it must remain possible to be disassembled without damaging it in order to support assembly, maintenance and component replacement [S3] .

For this purpose, a modular brick housing structure is proposed which consists of three separate parts:

- A supporting frame incorporating the docking face
- Interchangeable side-wall modules containing the receiving faces
- A final cover that closes the housing

This structure offers two main advantages: It allows the electronic components of the individual connection faces to be mounted separately before the modules are assembled into a complete building brick [S3] . As the receiving-side modules follow the same design regardless of their

position or the brick type, they can also be installed interchangeably and replaced individually when necessary.

5.4.3 Mechanical Concept of the Baseplate

The baseplate forms the starting point of every physical construction and consists of an enclosed housing for its technical components and one receiving face whose mechanical geometry corresponds to that of the building bricks. The first brick is placed on the baseplate using its defined docking face, which establishes it spatially and logically as the first building brick of the construction.

Additional receiving faces could enable several constructions with multiple starting points but would require an extension of the digital structure model and additional rules for comparison with the target structure. Consequently, the basic concept is limited to a single connection point.

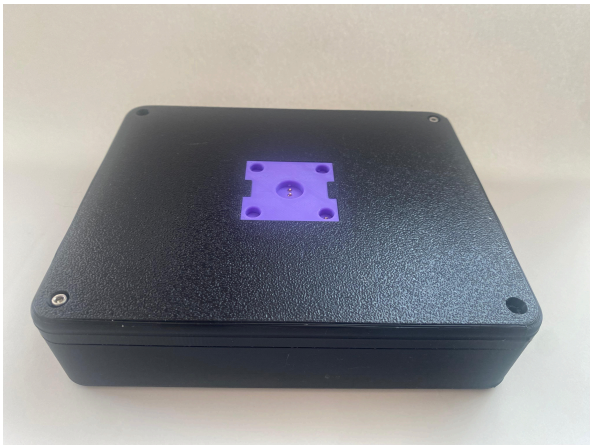

Figure 8: Baseplate with its single receiving face

5.5 Electronic Detection and Power Subsystem

Once the building bricks are physically connected, their arrangement must be made accessible to the technical system. The electronic design determines how reliably changes in the construction can be detected while keeping the active brick system operational. This section therefore describes the electronic foundation for sensing and supplying the physical construction. In accordance with the terminology defined in Section 5.3, the term upstream interface refers to the electronic interface on the docking face, whereas downstream interface refers to the electronic interface located on each receiving face.

5.5.1 Selection of the Electronic System Architecture

The selected electronic architecture results from the information that must be obtained directly from the physical construction: Each brick must be identifiable, while its connected face and relative rotation must be detected locally in order to reconstruct branching three-dimensional structures [DET1] [DET2] [DET3] . In addition, feedback must enable learners to identify which part of the construction requires attention and must therefore be clearly linked to the affected physical brick [FB2] .

Passive bricks would reduce the electronic complexity of the individual components, but their state would have to be determined entirely by external sensing infrastructure. Camera-based approaches depend on an unobstructed view of the construction and may require calibration to the interaction area, making them susceptible to occlusion by other bricks or the learner's hands [33]. The reviewed capacitive and optical approaches are tied to a sensing surface and limited to a detectable stack height [30, 34]. The construction systems based on **RFID** likewise support only vertical stacks and a limited stacking number. In addition, the UHF readers employed by systems

such as *RFIBricks* can be comparatively expensive, large and power-intensive [28, 29], making them unsuitable for use in schools and conflicting particularly with calibration-free use [IA2] .

An active architecture addresses these limitations by moving the acquisition of connection information to the individual bricks. Local electronics allow each brick to provide its identity, evaluate the state of its own interfaces and control a local feedback output. The information required for reconstructing the construction can therefore be obtained at the physical connection itself rather than inferred entirely through external sensing. This direct acquisition of connection data and the ability to address individual physical bricks provide the basis for selecting active bricks [32, 38].

Communication between the active bricks is established through their physical connections rather than through a separate wireless network. The problem with wireless transmission of an identifier, as used by *E-Block*, is that it would not by itself indicate which faces are connected or how neighbouring bricks are oriented [37]. Since electrical contacts are already required to distinguish the relative rotation at each connection, the same interfaces can also be used to transmit the resulting data. This avoids an additional wireless communication layer and creates a direct correspondence between the physical construction and its electronic connections. The resulting system is thus based on active, contact-connected bricks, drawing on principles found in *i-Cube*, *roBlocks* and *ActiveCube* [31, 32, 38].

5.5.2 Centralised Power Supply

In accordance with [EL1] , the building bricks must operate without individual batteries or rechargeable cells to reduce maintenance and decrease the necessary space within each brick. The active-brick systems of Watanabe et al. (2004) and Schweikardt and Gross (2006) address this requirement through a dedicated central power component [31, 32]. Following these approaches, the baseplate, which already forms the fixed starting point of the proposed construction, is also used as its central power-distribution point. It supplies the connected bricks through their electrical interfaces, while each brick uses the incoming power for its own electronics and forwards it through its downstream interfaces to subsequent child bricks.

5.5.3 Functional Structure

The building bricks and the baseplate share several fundamental electronic functions, including control, electrical interfacing and power distribution. However, these functions are distributed differently between the decentralised bricks and the central baseplate. Figure 9 presents the resulting functional structure and the interfaces between the two component types.

Each building brick acts as a local node within the physical construction, with its control unit maintaining the brick's identity and type while processing the signals received through its electrical interfaces. The upstream interface connects the brick to its parent, whereas a type-dependent number of downstream interfaces enables child bricks to be attached. Power received from the parent supplies the local electronics and is subsequently forwarded through the downstream interfaces. In addition, an integrated visual output allows feedback to be presented directly on the affected brick.

In contrast to the distributed functions of the individual bricks, the baseplate assumes the central role within the electronic subsystem. Through its single downstream interface, it connects to the first brick, initiates the detection process and supplies the resulting construction with power. At the same time, its application interface mediates communication between the physical brick system and the learning application. Since the baseplate forms the root of the construction, it does not require an upstream interface.

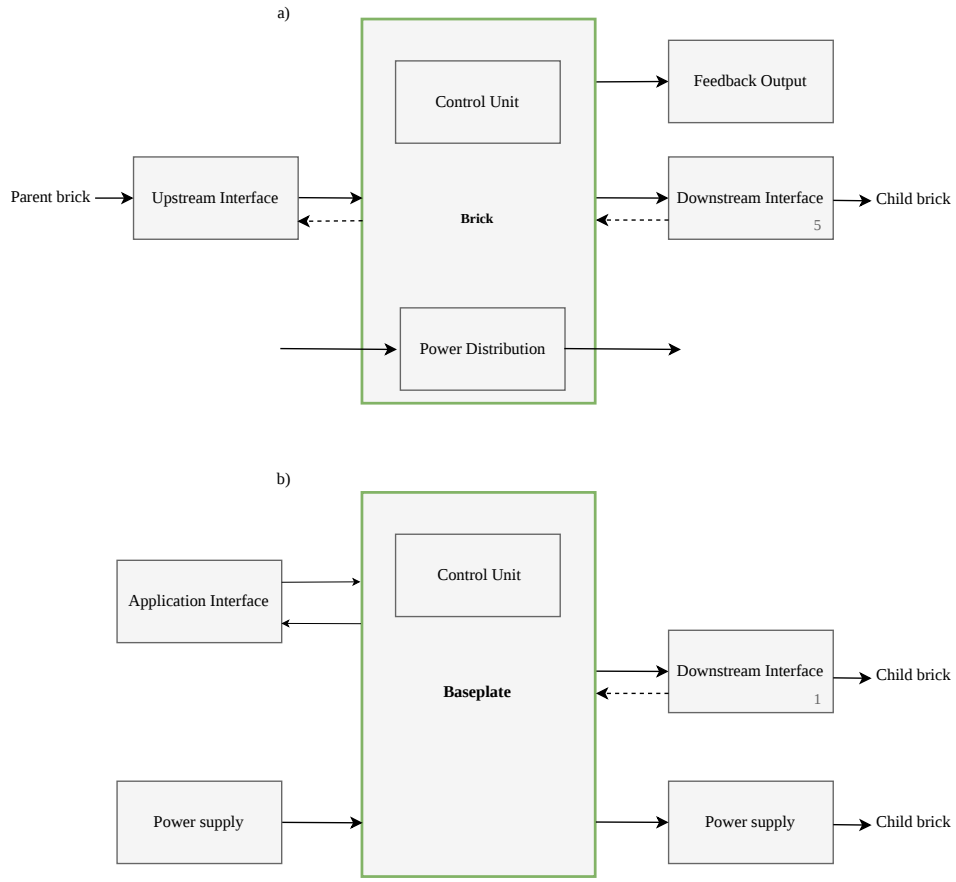

Figure 9: Functional structure of (a) a brick and (b) the baseplate

5.5.4 Electrical Interfaces and Local Detection Principle

Each mechanically defined connection is complemented by an electrical interface. Both interface types contain contacts for the power supply and the shared ground potential, but they differ in the arrangement and function of their signal contacts. The principle used for determining relative rotation is adapted from *i-Cube*, which distinguishes four orientations through a symmetrical fourfold sensor arrangement [38]. The proposed system transfers this principle to its directed electrical contact arrangement. Accordingly, the docking face of each brick contains four upstream orientation contacts positioned at its corners, with each contact representing one of the four mechanically permissible rotations. Each receiving face, by contrast, contains a single downstream child contact. Its position ensures that it connects to exactly one of the four upstream orientation contacts when two bricks are joined. Unlike *i-Cube*, the proposed arrangement therefore assigns different contact functions to the docking and receiving faces rather than providing the same transmitter-and-receiver arrangement on every face. The power-supply and ground contacts are arranged rotationally symmetrically so that they remain connected in all four permissible orientations. Figure 10 illustrates the resulting contact arrangements.

In support of [S4] , the recessed interface regions and mechanical guiding elements align and protect the electrical contacts, thereby reducing the risk of incorrect contacting and accidental short circuits.

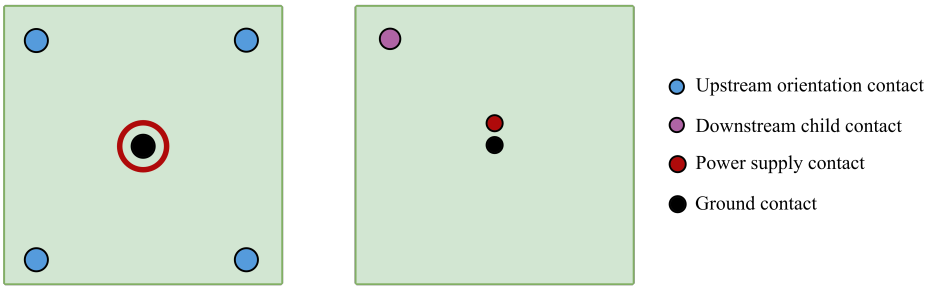

Figure 10: Contact arrangement of (a) the upstream interface and (b) the downstream interface

Detection of the Connection Face and Relative Rotation

When a child brick is attached, the downstream contact on the receiving face of the parent connects to one of the four upstream orientation contacts on the child's docking face. Because each receiving face has a separate downstream interface, the parent can query its interfaces individually and thereby determine through which of its faces the child is connected [DET2] . At the same time, the upstream contact activated on the child identifies which of the four relative rotations has been assumed [DET3] . The combination of the queried downstream interface and the contacted upstream orientation contact therefore defines the local spatial relationship between the two bricks. Once the connection has been detected, the same signal path is used for communication between the parent and child.

This information describes only the position and orientation of the child relative to its immediate parent. The electronic subsystem does not determine the global pose of an individual brick within the complete construction. Instead, the learning application derives the global positions and orientations by combining all locally detected parent–child relationships, as described in Section 5.6.4.

5.5.5 Local Electronic Feedback Output

The active-brick systems reviewed in the related work demonstrate how integrated visual outputs can communicate system states directly through physical bricks [31, 32, 37, 38]. Drawing on this principle, each building brick in the proposed system contains a simple visual feedback output, allowing feedback to be assigned directly to the affected physical brick rather than being presented exclusively through the learning application. The meaning, presentation and coordination of local and global feedback are specified in Section 5.8.5.

5.6 Data, Communication and Software Subsystem

The locally detected connection information must be combined into a coherent representation before the physical construction can be interpreted by the learning application. This transformation links the active brick system with the digital processing of the construction task. The following section presents the communication and software concept underlying this process and uses the terms parent node and child node to describe the directed relationships between connected bricks within the digital tree.

5.6.1 Overall Data Flow

Figure Figure 11 summarises the data flow between the learning application, the baseplate and the active brick system. After a target structure has been selected and stored by the learning application, the application sends a start signal to the baseplate. The baseplate then initiates recurring detection cycles in which the bricks recursively acquire their local connections and combine them into a serialised representation of the complete construction tree. This representation is forwarded

unchanged to the learning application, where it is deserialised, spatially reconstructed, compared with the target structure and translated into visual and brick-specific feedback. The feedback message is returned through the baseplate to the affected bricks before the next detection cycle begins. When the task is completed or terminated, the learning application sends a stop signal. The baseplate then terminates the detection process and waits for a new start signal.

This data flow separates physical acquisition from task-related interpretation. The baseplate and brick system detect and transmit the physical construction without knowledge of the selected target, whereas the learning application manages the task, interprets the detected structure and generates the corresponding feedback.

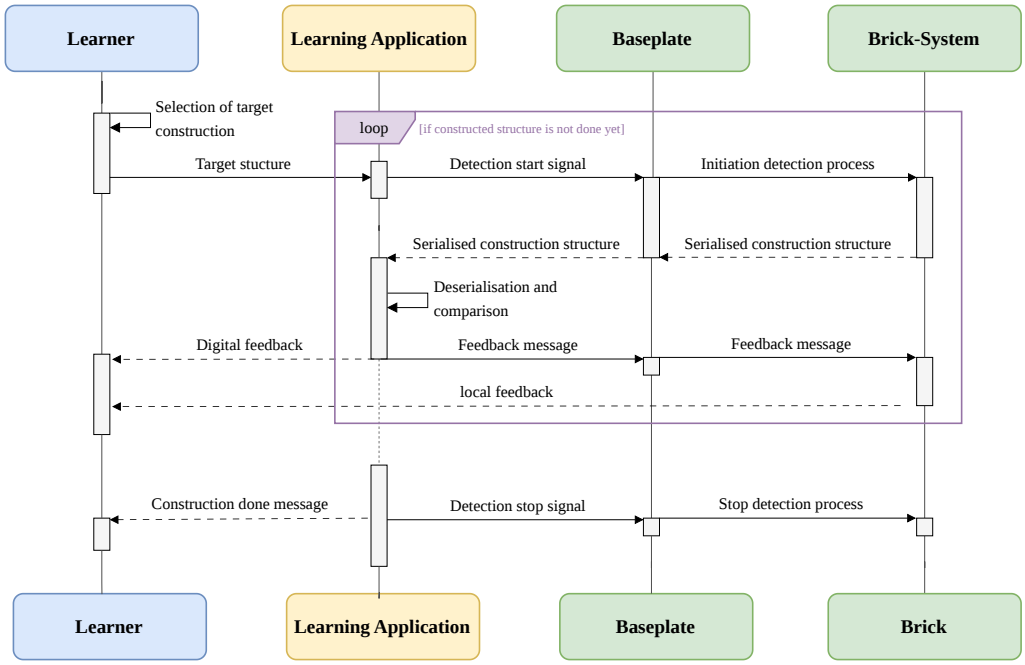

Figure 11: Overall data flow during a construction task

5.6.2 Logical Structure and Data Model

The physical construction is conceptually represented as a directed tree: The baseplate forms the root of the tree, each connected building brick is represented by a node and each physical connection constitutes a directed edge between a parent and a child node. The model also reflects the directed interface structure introduced in Section 5.5.4: The defined docking face connects each brick to its parent, while its receiving faces allow one or more child bricks to be attached. Consequently, each brick can have no more than one parent but may have several children. As required by [C1] , both linear and branching three-dimensional constructions can therefore be represented by the model. By deriving the digital interpretation from the parent–child relationships and the resulting overall construction, the structure model applies the relational and constructive mapping principles introduced in Section 2.4.2.3.

To identify and interpret each building brick within this tree, however, the parent–child relationship alone is insufficient. Table 13 summarises the additional information contained in each node for this purpose:

Information	Meaning
Identity	Unique assignment of the node to a physical building brick
Brick type	Geometric and functional properties of the building brick
Relative rotation	Orientation of the building brick in relation to its parent
Occupied connection faces	Assignment of child nodes to the corresponding receiving faces
Child nodes	Node information of the directly connected child bricks and their subordinate structures

Table 13: *Node information*

In accordance with [DET1] , the identity uniquely associates each node with an individual physical building brick and subsequently allows local feedback to be assigned to that brick. The brick type is required to interpret its geometry and compare it with the corresponding element of the target structure. Finally, the relative rotation and occupied connection faces describe the local spatial relationship between the node and its parent and children.

5.6.3 Detection and Communication Concept

Communication within the physical construction follows the parent–child relationships represented by the logical structure model in Section 5.6.2. Similar to the Comm-Block component of *roBlocks*, the baseplate acts as the central coordinator and initiator of each detection cycle [31]. It starts the acquisition of the construction by querying the first connected brick, after which communication proceeds recursively through the subordinate branches. Communication between two connected bricks follows a request–response principle. The parent brick requests information from a directly connected child brick, which responds with its own node information and the serialised representations of its subordinate structures. This communication takes place through the upstream interface of the child brick and the corresponding downstream interface of the parent brick described in Section 5.5.4. When the child brick subsequently queries its own receiving faces, it assumes the role of a parent brick in relation to the bricks connected below it.

Since only one signal path is available between the downstream child contact of the parent brick and the contacted upstream orientation contact of the child brick, the line must be used alternately for both transmission directions. The communication concept consequently employs a bidirectional single-wire half-duplex protocol. The exact message format, timing parameters and protocol implementation are described in the realisation section Section 6.2.

Sequential Interface Query and Recursive Detection

During each detection cycle, the downstream interfaces of a building brick are examined sequentially in a predefined order, following the round-robin query principle used in *i-Cube* [38]. This ensures that every available receiving face is checked regularly. If no child brick is detected at an interface, the brick continues with its next downstream interface. If a child brick is detected, its complete subordinate structure is acquired before the parent brick proceeds to the next interface. The overall construction is therefore detected using a depth-first search. Starting from the baseplate as the root, the system follows a detected branch to its outermost brick before processing the next downstream interface of the preceding parent brick. Once all downstream interfaces of a brick have been queried, the brick combines its own node information with the collected representations of its child nodes. The resulting representation of the complete subtree is then returned to the parent brick.

This procedure is applied recursively until the baseplate receives a serialised representation of the entire construction from the first connected brick. Each brick therefore processes only its local

connections, while the representation of the complete structure emerges through the aggregation of the individual subtrees.

Serialisation of the Tree Structure

A pre-order traversal is used to serialise the detected tree. The information of the current parent node is placed before the serialised representations of its child nodes. The child nodes are appended according to the predefined order in which the downstream interfaces of the corresponding parent brick are queried. Together with the information about occupied receiving faces, this consistent ordering enables the learning application to assign each deserialised child node to the correct connection face.

The sequence in which requests propagate through the physical construction must be distinguished from the ordering of the data within the resulting representation. Detection requests travel from the baseplate towards the outer bricks, whereas the collected subtree data return in the opposite direction. Within each serialised subtree, however, the parent node precedes its child nodes in accordance with the pre-order traversal. Figure 12 illustrates this distinction using an example physical construction and its corresponding tree model. The resulting serialisation order is indicated by the alphabetical order of the bricks; for clarity, the relative rotations are not shown.

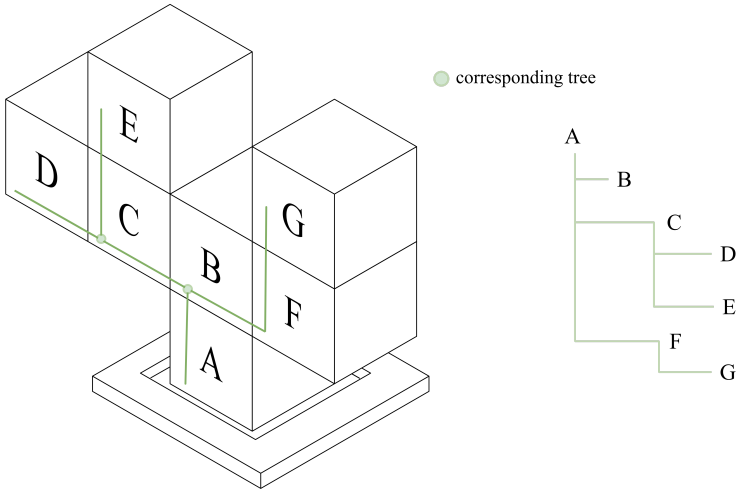

Figure 12: Physical construction and corresponding digital tree

5.6.4 Processing in the Learning Application

The baseplate transmits the serialised description of the detected construction to the learning application. The application deserialises the data and restores the individual nodes and their parent–child relationships as a digital tree. The predefined order of the downstream interfaces, together with the occupied connection-face information contained in each node, enables the child nodes to be assigned to the corresponding receiving faces in accordance with [DET2] . Using the baseplate as the origin of a shared coordinate system, the learning application subsequently calculates the global position and orientation of each building brick. For every parent–child connection, a local transformation is derived from three elements:

- The receiving face used for the connection
- The types of the two connected building bricks
- The detected relative rotation

The global pose of a child brick is obtained by combining the global pose of its parent with this local transformation. By applying this procedure recursively to all nodes, the complete physical construction can be reconstructed within the shared coordinate system. The resulting spatial model

forms the basis for the three-dimensional visualisation and the subsequent comparison with the selected target structure.

Comparison with the Target Structure

In accordance with [DET4] , the learning application compares the reconstructed current structure with the selected target structure. As already mentioned, structural correctness is determined by the brick types and their spatial relationships rather than by their individual identities. Bricks of the same type can therefore be exchanged without affecting the correctness of the construction. The comparison determines whether the current structure:

- Corresponds to a permissible partial structure of the target
- Completely matches the target structure, or
- Contains a connection, brick type, orientation, or position that deviates from the target

5.6.5 Feedback Channel

Based on the comparison with the target, the learning application calculates a feedback status for the overall structure as well as for the individual building bricks. The reconstructed digital construction and the global feedback are displayed directly in the learning application.

For local feedback, the relevant physical bricks are addressed through their identities. The learning application combines the brick-specific feedback states into a feedback frame and sends it to the baseplate, which forwards the feedback information through all the connected bricks. Each brick then checks whether its identity is part of the feedback frame and, if so, activates its local visual output. The pedagogical meaning and visual presentation of the individual feedback states are defined separately in Section 5.8.

5.7 Task and Interaction Subsystem

Once the physical construction has been reconstructed digitally, it must be embedded in a task structure that determines how learners can interact with it and how their progress is interpreted. The task and interaction subsystem therefore defines how the reconstruction task combines clear evaluation criteria with sufficient flexibility for learners to explore, revise and correct their construction process.

5.7.1 Physical–Digital Coupling

The interaction concept combines the static and interactive forms of coupling introduced in Section 2.4.1.2. At the representational level, each physical brick type is statically linked to a corresponding digital model shown in Figure 13. The cube brick and the smashed brick are therefore always represented by digital models that reflect their respective physical geometry and appearance. The meaning of an individual brick within a construction, however, is determined interactively. Its position and orientation, its role within the parent–child structure and its evaluation in relation to the target structure depend on the current physical configuration. Whenever bricks are connected, detached, rotated or repositioned, these properties are recalculated by the learning application. The digital meaning of the construction therefore emerges from the relationships between the physical bricks rather than from the individual elements in isolation.

5.7.2 Task Logic

The construction task is governed by a set of structural and procedural rules.

Following each detected change, the current construction is compared with the selected target structure. A partial construction is considered valid if its building brick type and connections form part of the target structure. The comparison therefore considers the type, the connection face used

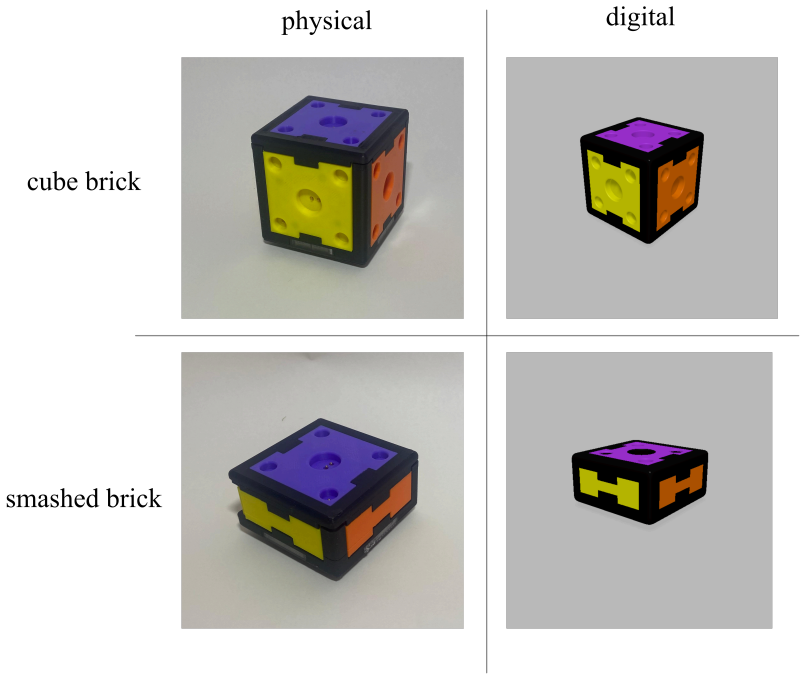

Figure 13: Comparison of the physical and digital building blocks

and the spatial position and the orientation of each brick. Building bricks of the same type are interchangeable within the construction task.

The individual identity of a brick is not relevant to the correctness of the solution but is required only for technical identification and for assigning local feedback to the corresponding physical brick.

The construction sequence is not prescribed in full. Different sequences may lead to the same valid result, thereby providing the openness required by [TUI1] . However, the directed parent–child structure imposes a necessary dependency: a child brick can only be attached after its parent is present in the physical construction.

The task is completed once the detected current construction fully corresponds to the target structure in terms of the brick types, connections, positions and orientations.

5.7.3 Available Actions and Error Correction

Within the defined task objective, learners can explore different construction paths and revise previously completed steps. Connected bricks can therefore be detached from the construction, rotated, attached to a different receiving face or replaced by another brick of the correct type. As required by [IA5] , errors are explicitly permitted and incorrect actions neither terminate the task nor result in other penalties. Since the physical construction remains visible, inspectable and directly modifiable throughout the task, the interaction also provides the structured openness required by [TUI1] .

5.7.4 Adaptation of Task Difficulty

According to [TUI2] , the difficulty of the construction tasks must be adaptable to different learning prerequisites and can be varied through the following characteristics:

- Number of building bricks: Tasks with fewer bricks provide a simpler starting point, whereas a larger number enables more extensive constructions.
- Spatial extent: Structures may extend along one, two, or all three spatial axes.

- Structural complexity: Tasks may involve linear constructions or branching structures with several child bricks.
- Relative rotation: Blocks may use consistent orientations or different rotations in increments of 90°.
- Block types: A task may require only one brick type or a combination of different types.
- Presentation of the target: The complete target structure may be displayed from the beginning or revealed incrementally.
- Level of support: Feedback may be limited to an evaluation of the current result or supplemented with guidance concerning possible subsequent actions.

These characteristics can be combined to create tasks of varying complexity without changing the fundamental interaction principle.

5.8 Visualisation and Feedback Subsystem

The evaluation produced by the task and interaction subsystem must be made perceptible to learners so that they can relate the system response to their physical actions. The following sections translate the detected construction and the results of its comparison with the target into coordinated outputs on the physical bricks and within the learning application.

5.8.1 Visualisation of the Target Construction

At the beginning of a construction task, a target structure is selected in the learning application. This structure remains available as a visual reference throughout the construction process and can be presented either completely or incrementally. A complete representation enables learners to analyse the target independently and plan their own construction process. An incremental representation provides additional guidance by presenting or highlighting only the currently relevant part of the target structure. As required by [TUI2] and [FB4] , different levels of difficulty and support can therefore use different target-structure visualisations.

5.8.2 Visualisation of the Current Construction

Following each detection cycle, the learning application represents the reconstructed current structure as a three-dimensional model. Each detected building brick is positioned within the scene according to its type, calculated position and orientation. Its digital representation reflects the geometry and appearance of the corresponding physical brick to maintain a comprehensible relationship between the physical and digital representations as required by [TUI3] .

The current structure is presented in an interactive three-dimensional view that supports rotation and zooming, allowing the construction to be inspected from different perspectives. The visualisation remains available throughout the task, while the physical construction remains directly accessible and can be modified at any time. Together, these properties are intended to keep the resulting construction visible, inspectable, and modifiable throughout the task [IA1] . This is particularly relevant for branching structures in which individual bricks or spatial relationships may be obscured from a single viewpoint. When the visualisation is initially displayed, the camera is automatically aligned with the defined front view of the construction along the positive y-axis. The target structure and the detected current structure must remain visually distinguishable to prevent learners from confusing the reference with their own construction. They are therefore displayed in separate views or clearly separated spatially within the application.

5.8.3 Integration of Visualisation and Feedback

The representation of the current structure is supplemented with its brick-specific feedback derived from the comparison with the target. While the relationship between a physical brick type and its digital geometry remains statically defined, the feedback state assigned to a brick changes dynam-

ically according to its current position within the construction (see Section 2.4.1.2). This feedback visualisation distinguishes four relevant states:

- A detected brick that is correctly positioned
- A detected brick that deviates from the target structure
- An unoccupied position that constitutes a permissible next construction step
- A target position that is not yet accessible because its required parent brick is missing

To make the different states visible to the learners, different coloured highlights are used for the building bricks. Correct and deviating bricks are distinguished through contrasting visual highlights, such as green and red outlines. Permissible next positions can be represented as transparent placeholders or *ghost cubes*. Positions that are not yet accessible are not displayed, as their presentation could suggest an action that cannot yet be performed within the directed parent–child structure. By attaching each visual state directly to the affected brick or position, the feedback can be clearly associated with the relevant part of the construction in accordance with [FB2] .

5.8.4 Feedback Levels

The concept combines three levels of feedback and draws on the different feedback modalities introduced in Section 2.5.1. Intrinsic physical feedback results from the mechanical guiding elements and magnetic assistance. When connecting two bricks, learners can physically perceive that the two faces have engaged. However, this confirms only the mechanical connection and does not indicate whether the connection has been detected electronically or whether the brick is correct in relation to the target structure. It is therefore supplemented by digital feedback. Its local form is presented directly on the affected physical brick, on which a green or red light indicates whether the connected brick has been placed correctly or needs to be reviewed. The global digital feedback is presented through the learning application. In addition to the state of individual bricks, it can represent the complete construction, the progress made towards the target and possible subsequent actions. It therefore supports more detailed information than can be communicated through the limited output capabilities of an individual brick.

The local and global outputs must remain semantically consistent: if a brick is marked as correct locally, the corresponding digital brick must display the same state in the learning application.

5.8.5 Types of Feedback

Based on the distinction established in the theoretical background in Section 2.5.2, the concept differentiates between interaction feedback and pedagogical feedback, as required by [FB1] .

By showing which digital structure the system has derived from the physical construction, the visualisation provides interaction feedback: The learner can infer that a connection has been detected and processed if an added brick appears at the position and orientation in which it was physically placed. If the brick is absent or appears differently from the physical arrangement, this indicates a possible detection or reconstruction problem.

The pedagogical feedback evaluates the detected state in relation to the target structure and can fulfil three functions:

- Outcome feedback indicates whether a brick or connection is correct
- Progress feedback shows how much of the target structure has already been completed
- Feedforward indicates a permissible subsequent action or position

Outcome and progress feedback constitute the basic feedback provided by the prototype, while feedforward can supplement these forms depending on the configured level of support. Table Table 14 summarises the principal feedback cases and their presentation at the local and global levels.

Feedback case	Local feedback	Visualisation in the learning application	Type of feedback
Physical change detected	No separate local output	The detected current construction is updated to show the recognised brick, position and orientation	Interaction feedback
Connection detected and correct	Positive light signal on the affected brick	The brick is visually highlighted as correct	Outcome feedback
Connection detected but inconsistent with the target	Negative light signal on the affected brick	The brick or affected area is visually highlighted as deviating	Outcome feedback
Permissible next position	No local output	Optional representation as a transparent ghost cube	Feedforward
Partial progress achieved	No local output	The progress indicator is updated	Progress feedback
Target structure completed	No local output	Successful completion of the task is displayed	Outcome feedback

Table 14: *Feedback cases and their presentation*

Graduated Support

Different types of feedback should be provided at different points in time: Interaction feedback should be provided promptly after the action that triggered it, as required by [DET7] , so that learners can associate the system response with the corresponding physical connection. Outcome feedback should likewise indicate promptly whether the detected construction step corresponds to the target structure.

According to [FB3] as well as [FB4] , more extensive assistance is provided incrementally according to the learner's support needs. To allow the learner to inspect the physical and digital construction and attempt an independent correction, only the affected brick is initially marked as deviating. If further support is required, the learning application may provide an additional explanation. At a higher level of assistance, permissible next positions can be displayed as ghost cubes to provide feedforward. As already mentioned, the representation of the target structure can also be adapted to the required level of support. It may remain fully visible to facilitate independent planning or be presented incrementally to direct attention towards the next relevant part.

This graduated approach preserves opportunities for independent error analysis while providing additional orientation when required.

6 Technical Realisation

The following chapter describes how the concept of the proposed learning system was implemented in the physical prototype. The focus is placed on the manufactured components, the electronic hardware, the implemented communication protocol, the embedded firmware and the learning application.

6.1 Mechanical and Electronic Realisation

The physical and electronic implementation of the building bricks and the baseplate is described here. The section focuses on the components and construction methods used in the prototype as well as on deviations that occurred during manufacturing.

6.1.1 Mechanical Implementation

The prototype comprises two physically implemented brick types: the *cube Brick* and the *smashed Brick*. Both types use the standardised square connection interface defined in Section 5.4.1 but differ in their geometry and number of receiving faces. In contrast to the cube brick, the smashed brick has only half its height and provides a single receiving face opposite its docking face. Its four remaining lateral faces do not permit further connections.

The square connection faces have a side length of approximately 40 mm. The selected dimension represents a compromise between the one-handed grasping and positioning required by [C3] and the internal space required for the electrical contacts and electronic components. The ergonomic suitability of the implemented dimensions is assessed in the evaluation chapter.

Both brick types were modelled using Autodesk Inventor and manufactured using a 3D printer. Their housings implement the modular structure introduced in the concept chapter and consist of a black structural frame, colour-coded connection-side modules and a removable black cover. The side modules are inserted individually into the supporting frame and can be removed separately. This allows the electronic components to be mounted and tested before the complete brick is assembled. In addition, individual modules can also be replaced without manufacturing an entirely new housing.

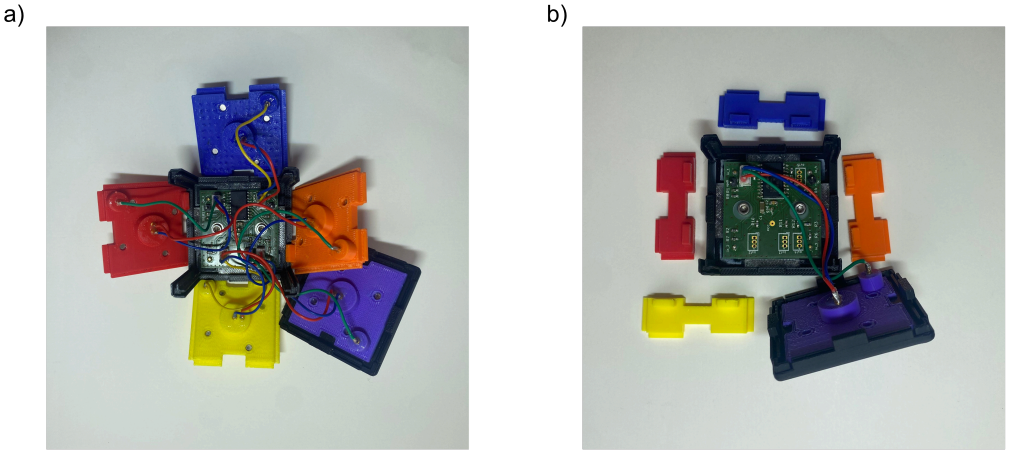

Figure 14: Exploded views of the prototype housings: (a) cube brick and (b) smashed brick

The baseplate is implemented as the root element of the construction. Its housing contains the baseplate electronics and provides one receiving side for connecting the first brick.

6.1.2 Electronic Implementation

Both brick types are equipped with an ATtiny1616 microcontroller that serves as their local control unit. The mechanical interfaces contain spring-loaded pogo pins that establish the electrical connections while compensating for minor manufacturing tolerances and small movements between connected bricks. The recessed guiding structures surrounding the contacts additionally protect the electrical interface against accidental contact, incorrect alignment and short circuits. The electronic structures of the building bricks and the baseplate are summarised in Figures Figure 1 and Figure 2.

The electrical implementation differs between the docking and receiving faces. On a receiving face, the data, supply and ground connections are implemented using spring-loaded contacts. The docking face contains the corresponding contact surfaces for the data connections, whereas the power connection uses a different construction consisting of a neodymium ring and a pogo pin mounted directly on the printed circuit board. [Figure to be added]

The four upstream data contacts of the docking face are located in the corners of the square connection area. Ground and power of this face are transmitted through rotationally symmetrical contacts. The ground contact is positioned at the centre of the interface, while the supply contact forms a concentric connection around it. Both electrical connections therefore remain unchanged when a brick is rotated in increments of 90°. A receiving face provides a single downstream data contact in one defined corner as well as the power and ground contacts in the centre. Because of the limited installation space, the downstream data contact of one receiving face of the cube brick is placed in a different corner from the otherwise uniform arrangement. The resulting rotational offset is compensated for in the brick firmware, as described in Section 6.3.1.2. The implemented contact arrangements are shown in Figure 15.

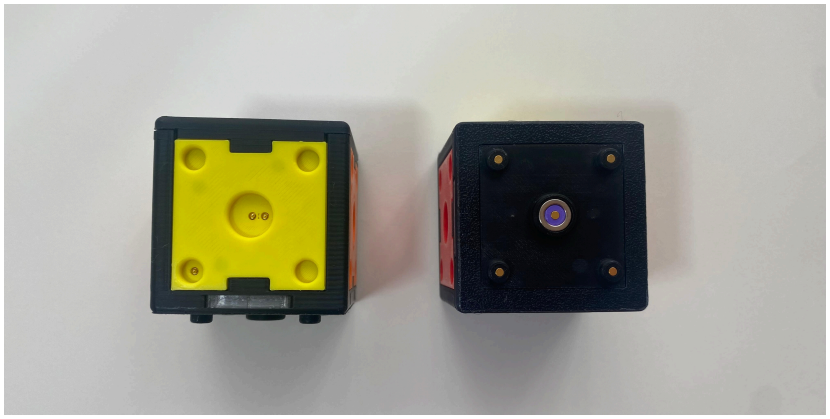

Figure 15: Implemented contact arrangements: receiving face (left) and docking face (right)

Four side-view RGB LEDs are arranged around the printed circuit board to provide local feedback directly on each building brick. The ATtiny1616, the RGB LEDs and the electrical components of the docking face are mounted on a printed circuit board designed using KiCad. The PCB is attached to the inside of the docking face using screws, while the contacts of the receiving faces are connected to the main board through the brick's internal wiring. The complete circuit schematic and PCB layout are provided in the Appendix.

The baseplate is also controlled by an ATtiny1616. It communicates with the computer through the microcontroller's hardware USART interface and a connected USB-to-TTL UART adapter. The prototype is supplied with 5 V from the computer's USB interface through this adapter. The baseplate transfers the supply voltage to the connected root brick, from which it is forwarded through the subsequent brick connections.

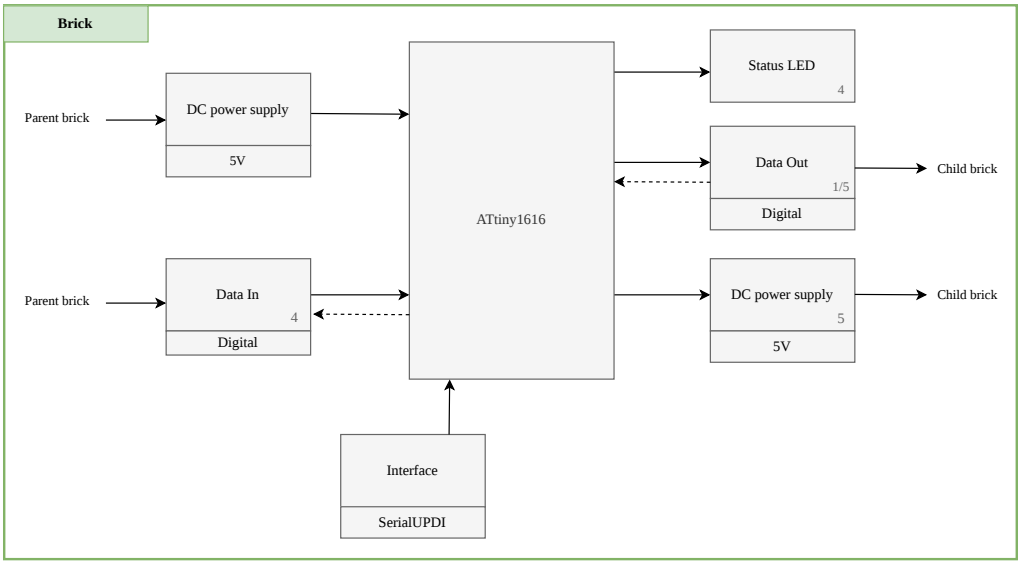

Figure 1: Electronic block diagram of a building brick

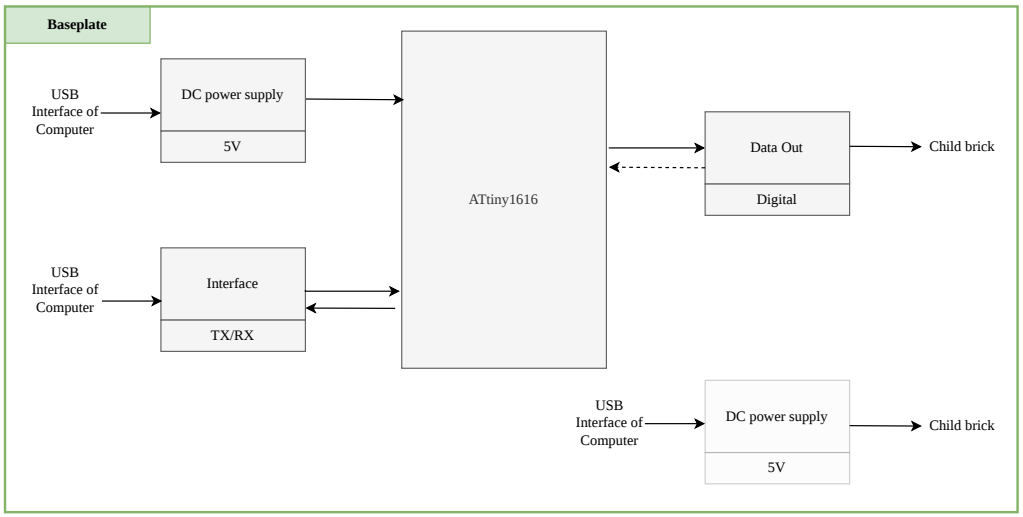

Figure 2: Electronic block diagram of the baseplate

6.2 Communication Protocol

The communication concept introduced in Section 5.6 is implemented using two separate communication interfaces. Communication within the physical construction follows the parent–child hierarchy and uses a custom, software-controlled single-wire half-duplex protocol. Communication between the learning application and the baseplate is implemented using the hardware USART of the baseplate microcontroller and a USB-to-TTL UART adapter. The two interfaces are summarised in Table 15.

Communication Interface	Physical Implementation	Transmission Rate
Learning application–baseplate	Hardware USART through a USB-to-TTL UART adapter	115200 bit/s
Baseplate–brick and brick–brick	Software-controlled single-wire half-duplex connection	approximately 9.6 kbit/s

Table 15: *Implemented communication interfaces*

6.2.1 Single-Wire Communication Between the Building Bricks

The communication between two directly connected modules is implemented as a bidirectional single-wire half-duplex interface. The same data line is used alternately by the parent and child brick and only one participant may transmit at a time.

The idle state of the line is logic high. However, the line is not actively driven high. Instead, the transmitting microcontroller either pulls the line to ground to transmit a logic low or releases it and activates its internal pull-up resistor to obtain a logic high. This open-drain-like behaviour reduces the risk of electrical contention because connected devices do not actively drive the line to opposing voltage levels. Each byte is transmitted using a software-implemented UART-like format illustrated in Figure 16.

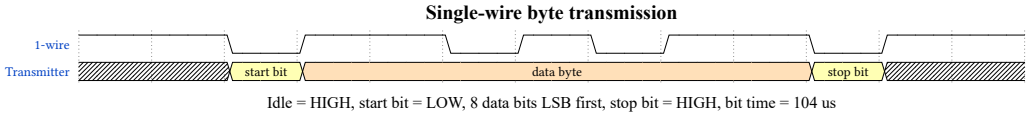

Figure 16: Timing of one byte on the single-wire interface

The logic-low start bit indicates the beginning of a byte. It is followed by eight data bits transmitted least significant bit first and a logic-high stop bit. No parity bit is used. A bit duration of 104 µs results in a transmission rate of approximately 9.6 kbit/s and a total transmission time of approximately 1.04 ms per byte. After a participant has completed a transmission, the line is released before the other participant responds. A turnaround interval of 312 µs, corresponding to three bit periods, is used whenever the direction of transmission changes. This communication type is used for construction detection and feedback-frame forwarding.

6.2.2 Common Frame Format

Construction and feedback data have different contents but are transmitted using the same variable-length binary frame format. This format, shown in Table 16, is used for communication between the building bricks as well as between the baseplate and the learning application:

SYNC	LEN	PAYLOAD	CHECKSUM
1 byte	1 byte	0-128 bytes	1 byte

Table 16: Implemented binary frame format

The SYNC synchronisation byte enables the receiver to identify the beginning of a frame. The length field given by LEN specifies the number of payload bytes that must subsequently be read. A frame with LEN = 0 contains an empty payload. Although the one-byte length field could theoretically represent up to 255 payload bytes, the prototype limits the payload to 128 bytes to restrict the required buffer size and transmission duration.

The payload stored in the PAYLOAD field contains the information transmitted by the frame. Its structure depends on whether it contains the detected tree structure or the feedback message. Both payload formats are described in detail in Section 6.2.3.1.1 and Section 6.2.3.2.1.

To detect transmission errors, an XOR CHECKSUM is calculated from the length field and all payload bytes:

$$C = L \oplus P_0 \oplus P_1 \oplus \cdots \oplus P_{L-1}$$

For an empty payload, the checksum is equal to the length field. Upon reception, the checksum is recalculated and compared with the transmitted value. If the two values do not match, the frame is discarded.

6.2.3 Parent–Child Communication

The communication within the physical construction follows a request–response procedure and is always initiated by the parent brick. Depending on the current process, the communication is used either to retrieve the detected construction or to transmit feedback towards the connected bricks.

The command and acknowledgement bytes used in the two processes are summarised in the following table Table 17.

Message	Value	Direction	Meaning
CMD_STATUS	0xAA	Parent ⟶ child	Request the detected subtree
ACK_BUSY	0x55	Child ⟶ parent	Request received and being processed
CMD_FEEDBACK	0xCC	Parent ⟶ child	Announce the transmission of a feedback frame
ACK_FINISHED	0x66	Child ⟶ parent	Feedback successfully processed and forwarded
ACK_FAILED	0x77	Child ⟶ parent	Feedback frame invalid or forwarding failed

Table 17: *Command and acknowledgement bytes of the parent–child protocol*

Detection Communication

To request the detected subtree of a connected child, the parent node sends CMD_STATUS through the corresponding downstream interface. After receiving this valid request, the child node changes the transmission direction and responds with ACK_BUSY. This confirms that the request has been received and that the child is beginning to collect the required information by subsequently querying its own receiving interfaces. Once the subordinate information has been collected, it returns a DATA_FRAME containing its own payload and the payload of its complete subtree. This procedure is recursively repeated at every occupied parent–child connection. Figure 17 illustrates one such communication transaction at byte level.

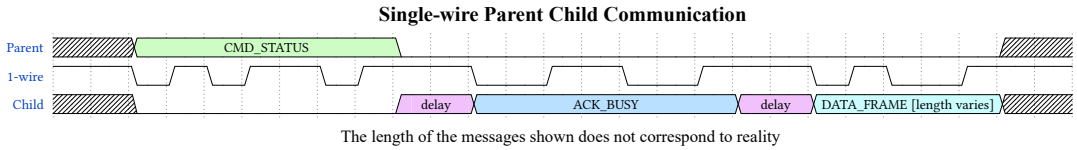

Figure 17: Byte-level timing diagram of the detection communication between a parent and child brick

6.2.3.1.1 Detection Payload

The payload returned during this sequence represents the detected construction tree. Each brick contributes a three-byte node header, followed by the serialised representations of its subordinate bricks shown in Table 18.

NODE_ID	ORIENTATION	CHILD_MASK	CHILD_NODE_PAYLOAD
1 byte	1 byte	1 byte	variable

Table 18: Detection payload

The three most significant bits of NODE_ID encode the brick type, whereas the five least significant bits encode the individual instance number. This allows the learning application to distinguish between the cube and smashed brick types while retaining an individual address for assigning local feedback. ORIENTATION contains a value from 0 to 3, representing the four possible rotations of the brick relative to its parent. Within CHILD_MASK, bit (i) indicates whether a child was detected at receiving interface (i). The cube brick can use bits 0 to 4 for its five receiving interfaces, whereas the smashed brick only needs one bit assigned to its single receiving interface. The node representations of the connected children follow the node header in ascending order of the receiving-interface indices. The precise procedure through which the firmware collects and combines these node representations is described in Section 6.3.1.

The baseplate does not add a node header of its own. Instead, it transfers the payload beginning with the first brick connected to it. If no root brick is detected, the baseplate transmits a DATA_FRAME with an empty payload.

Feedback Communication

The feedback communication is also initiated by the parent node, but with another command message. CMD_FEEDBACK signals to the child node the upcoming transmission of a FEEDBACK_FRAME. After receiving the command, the child responds with ACK_BUSY, indicating that it is ready to receive the frame. The parent then transmits the complete FEEDBACK_FRAME. After processing and forwarding the frame, the child reports the result using either ACK_FINISHED or ACK_FAILED. The internal processing and forwarding logic is described in Section 6.3.1.

6.2.3.2.1 Feedback Payload

The feedback payload uses the common frame format but contains pairs of brick identifiers and LED states:

NODE_ID	LED_STATE	NODE_ID	LED_STATE	…
1 byte	1 byte	1 byte	1 byte	…

Table 19: Feedback payload

LED_STATE can assume one of two values: LED_GREEN or LED_RED. If a brick's NODE_ID is not included in the feedback payload, that brick switches off its **LED**. An empty feedback payload can therefore be used to deactivate all local feedback simultaneously.

6.2.4 Communication between Learning Application and Baseplate

The connection between the learning application and the baseplate uses the ATtiny1616 hardware USART at 115200 bit/s with eight data bits, no parity and one stop bit. The learning application controls an interaction session using the following single-byte commands and responses:

Message	Value	Direction	Meaning
PC_CMD_START	0x53 (S)	Application → baseplate	Start cyclic construction detection
DATA_FRAME	Variable	baseplate → Application	Current serialised construction
FEEDBACK_FRAME	Variable	Application → baseplate	Feedback calculated by the application

Message	Value	Direction	Meaning
PC_RESP_FB_DONE	0x44 (D)	baseplate → Application	Feedback successfully forwarded
PC_RESP_FB_FAILED	0x4E (N)	baseplate → Application	Invalid feedback frame or forwarding failure
PC_CMD_STOP	0x58 (X)	Application → baseplate	Stop detection and switch off local feedback

Table 20: *Messages exchanged between the learning application and the baseplate*

The sequence diagram in Figure 18 represents the following communication principle: After receiving PC_CMD_START, the baseplate begins the detection process and transmits the detected construction to the learning application using the DATA_FRAME format defined above. After receiving the DATA_FRAME, the learning application answers with the FEEDBACK_FRAME calculated from the target comparison. The baseplate forwards this frame through the physical construction and responds with PC_RESP_FB_DONE if all detected bricks confirmed the feedback. If the received frame is invalid or at least one brick fails to confirm the feedback transmission, it responds with PC_RESP_FB_FAILED.

When the baseplate receives a PC_CMD_STOP from the learning application, it forwards an empty FEEDBACK_FRAME so that all local LEDs are switched off. It then terminates the cyclic detection by returning to its idle state.

6.2.5 Validation and Error Handling

Several validation mechanisms were implemented to prevent incomplete or corrupted transmissions from being interpreted as valid construction data:

- Each received byte is checked for a valid start and stop bit.
- The SYNC byte allows the receiver to locate the beginning of a frame.
- The received length is checked against the maximum payload size of 128 bytes.
- Timeouts terminate the reception of incomplete bytes or frames.
- The XOR checksum detects changes to the payload or length field.
- On the computer interface, incomplete or invalid frames are discarded and reception resumes by searching for the next valid SYNC byte.

The baseplate waits up to two seconds for the first byte of the application's feedback response and allows a maximum interval of 100 ms between bytes within that frame. Detection cycles are currently separated by a delay of 550 ms, in addition to the time required for communication within the connected construction. These values describe the implemented prototype and are not general protocol requirements.

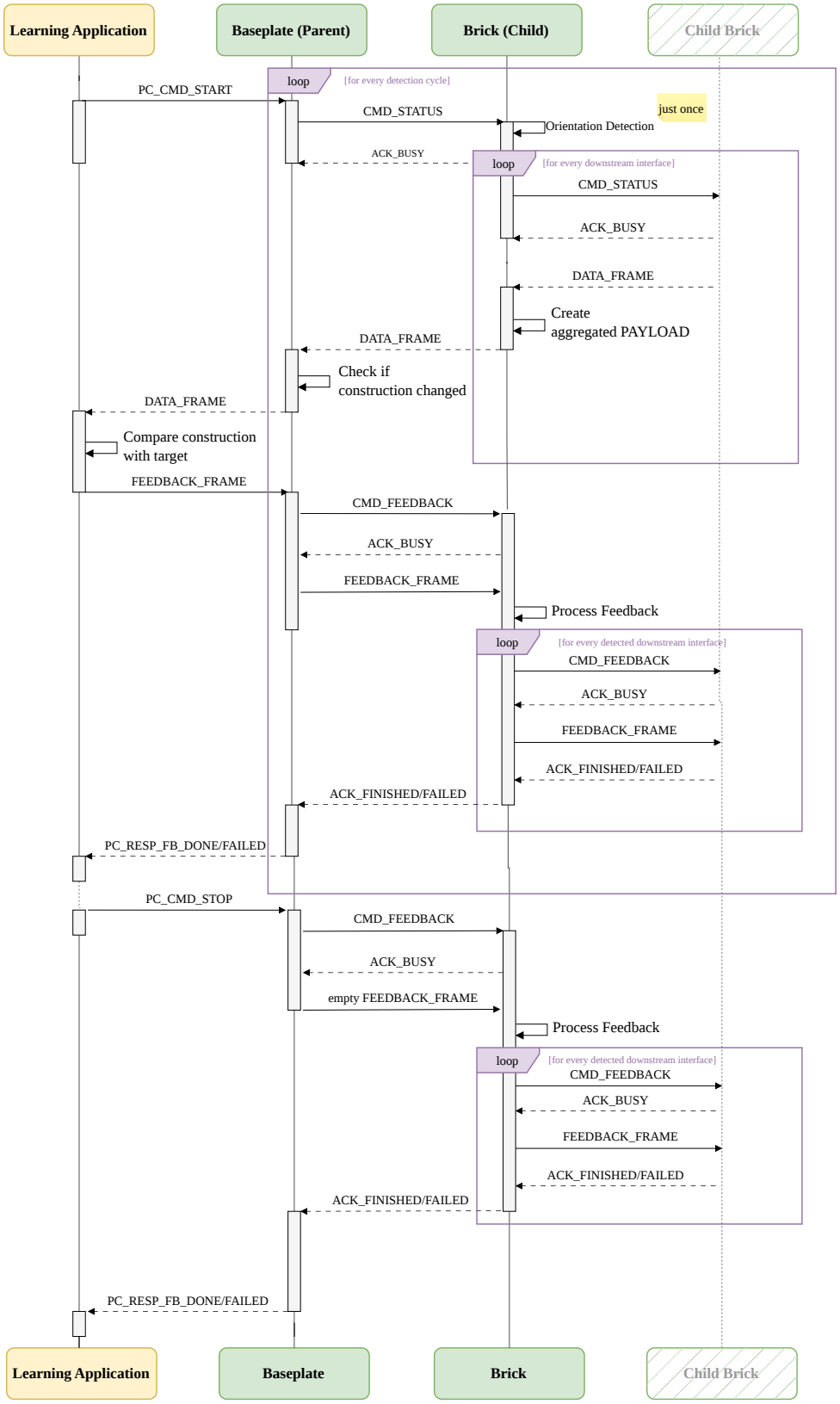

Figure 18: Communication sequence between the learning application, baseplate and connected bricks

6.3 Firmware of Bricks and Baseplate

To communicate changes in the physical construction to the learning application and return the resulting feedback to the relevant bricks, the distributed electronic components must coordinate their behaviour. The firmware of the bricks and the baseplate, explained in this chapter, implements this coordination by controlling construction detection, data transmission and feedback processing.

6.3.1 Brick Firmware

The brick firmware is responsible for collecting its local subtree and controlling the feedback output. The implemented control flow is shown in Figure 19. Part (a) presents the main processing sequence, while Part (b) details the initial detection of the upstream communication line and the brick's relative orientation.

As shown in Figure 19 (b), the firmware initially releases all communication lines and configures the **LED** outputs. It then monitors the four upstream contacts of the docking face. The contact through which the first valid request is received determines both the communication line used for subsequent exchanges with the parent and the relative orientation of the brick. This orientation detection is determined just once after the microcontroller has started. Disconnecting the brick interrupts its centrally supplied power, causing the firmware to restart when the brick is connected again. The orientation is therefore detected again after every new physical connection.

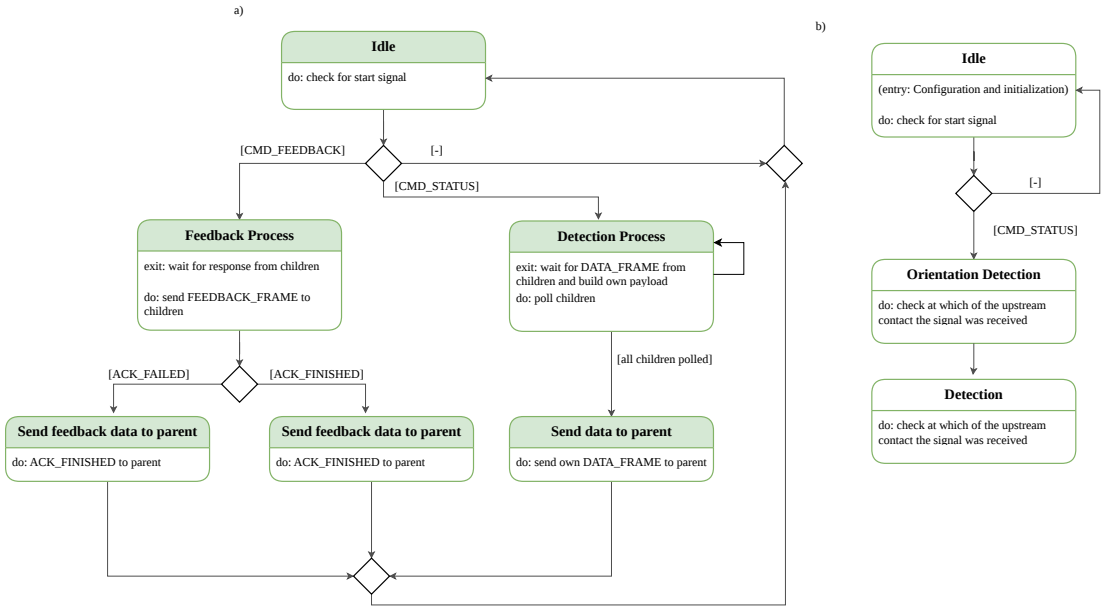

Figure 19: State machine of the brick firmware: (a) main processing sequence and (b) initial communication-line and orientation detection

Detection Process

When the state machine enters the detection state, it queries the available downstream interfaces in ascending order. Depending on the brick type, this means querying five interfaces or, for the smashed brick, only one. Communication with an individual child is handled by a dedicated child-polling function. A successful response provides the serialised payload of the child's complete subtree. This subtree payload is stored temporarily until all configured downstream interfaces have been processed. If a child does not return a valid payload within the available attempts, its subtree is omitted from the current detection result. Successfully received child-subtree payloads are appended in ascending interface order to the brick's own node payload according to the format specified in Section 6.2.3.1.1. For each included child, the corresponding bit in CHILD_MASK is set

to one. The resulting complete payload therefore consists of the current brick's own node payload followed by the payloads of its successfully detected child subtrees. It is then transmitted to the parent using the frame and message sequence defined in Section 6.2.

Because every child performs the same processing before returning its result, the complete construction is acquired recursively without requiring an individual brick to store a global representation of the structure.

Correction of the Contact Offset

As described in the mechanical realisation, the downstream data contact of one receiving face of the cube brick could not be placed in the same relative corner as the other downstream contacts. This causes the orientation reported by a child connected to this interface to contain an offset of one quarter turn. The brick firmware compensates for this deviation before it incorporates the child payload into the parent payload by incrementing the orientation value modulo four:

$$o_{\text{corrected}} = (o_{\text{detected}} + 1) \bmod 4$$

The learning application therefore does not require an additional interface specific correction because it consequently receives orientation values with a consistent interpretation for all receiving interfaces.

Feedback Processing

When the brick enters the feedback state, it receives and validates the FEEDBACK_FRAME according to Section 6.2.3.2.1. It searches the payload for its own NODE_ID and, if it is included, applies the corresponding state to the local LED output. If the NODE_ID is not part of the frame, the LEDs are switched off.

The same FEEDBACK_FRAME is subsequently forwarded to the interfaces marked as occupied during the preceding detection cycle. Using the messages defined in Section 6.2, the brick then reports to its parent whether the frame was successfully processed and forwarded.

6.3.2 Baseplate Firmware

The behaviour of the brick firmware is coordinated by the baseplate firmware, which initiates the detection and feedback processes. In addition to controlling the communication session, the baseplate mediates between the learning application and the physical brick structure. It does not parse the construction, calculate spatial positions or evaluate the task. These operations are performed by the learning application described in the following Section 6.4.

The implemented control flow is shown in Figure 20. After initialisation, the baseplate releases its downstream communication line, configures its hardware USART and enters the idle state. It remains in this state until the learning application starts a new communication session. During an active session, the baseplate repeatedly initiates the detection of the connected construction by querying the brick attached to its single receiving interface. Since this is the first brick of the construction, its response already contains the serialised payload of the complete root subtree. If no valid root payload is received, the current construction is represented by an empty payload. Otherwise, the received subtree payload is forwarded without structural modification to the learning application.

Payload Transmission Control

To avoid transmitting identical construction data during every cycle, specific conditions determine when the baseplate sends data to the learning application.

The first payload after the beginning of a session is always transmitted, irrespective of a potentially retained result from a preceding session. For every subsequent payload transmission, the baseplate compares the current detection payload with the payload transmitted most recently. A new DATA_FRAME is transmitted only when the current payload differs from the preceding

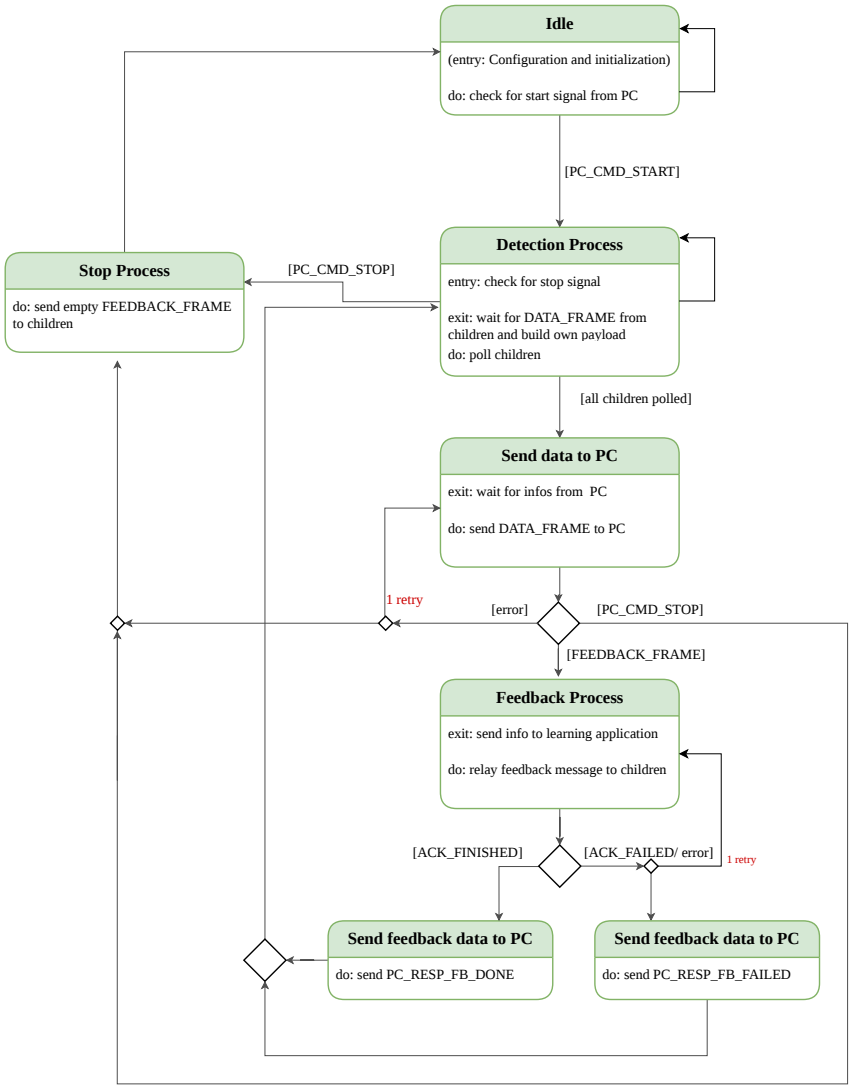

Figure 20: State machine of the baseplate firmware

one. If the construction remains unchanged, the baseplate sends the unchanged payload only after 50 detection cycles as a heartbeat. This confirms that the communication link and the detection process remain active without continuously transmitting identical frames.

The current implementation inserts a delay of 550 ms between consecutive detection cycles. The complete interval between two results additionally depends on the number and arrangement of connected bricks as well as on possible retransmission attempts.

Feedback Forwarding and Session Termination

After transmitting a DATA_FRAME, the baseplate waits for the corresponding FEEDBACK_FRAME from the learning application. It then forwards the frame to the root brick using the same procedure as the feedback processing of the building bricks described in Section 6.3.1.3. Afterwards, the baseplate reports to the learning application whether the frame was confirmed by the connected construction. The baseplate neither interprets the individual feedback entries nor controls the LEDs of the bricks directly. Its role is limited to receiving the frame from the learning application and initiating its distribution through the physical construction.

When the learning application terminates the session, the baseplate forwards an empty feedback frame to the root brick and since no brick NODE_IDs are contained in this payload, all local feedback outputs are switched off. The baseplate then ends cyclic detection and returns to the idle state shown in Figure 20.

6.4 Software Architecture and Backend Processing

In the current prototype, the learning application is executed on a host computer and consists of a Python-based backend and a browser-based frontend implemented in JavaScript. Both components run on the same computer. The backend processes the data exchanged with the baseplate, while the frontend provides the selected target structure and presents the evaluated construction to the user. This section describes the implemented software architecture and backend processing. The three-dimensional visualisation and user-facing presentation of feedback are addressed separately in Section 6.5.

6.4.1 Implemented Software Architecture

The backend acts as an intermediary between the browser frontend and the baseplate. It communicates with the frontend through a WebSocket connection using JSON objects, while PySerial is used to exchange the binary frames defined in Section 6.2 with the baseplate.

A construction session begins when the frontend transmits the selected target structure to the backend. For each construction received from the baseplate, the backend reconstructs the detected tree, compares it with the selected target and generates a feedback frame for the physical bricks as well as an evaluated tree for the frontend.

Serial communication is handled by a dedicated worker thread, whereas the WebSocket server operates asynchronously. The two interfaces exchange different data representations: compact binary frames are used for communication with the baseplate, while JSON objects are exchanged with the frontend. Since the prototype uses a single serial interface connected to one baseplate, only one browser-based construction session can be active at a time.

6.4.2 Deserialisation and Internal Tree Models

The backend receives the detected construction in the payload format defined in Section 6.2.3.1.1. A recursive parser converts this linear byte sequence into an object-based tree representation. Starting with the root node, the parser reads each node header and uses the transmitted CHILD_MASK to identify and recursively reconstruct its subordinate nodes in ascending port order. Each child node is thereby assigned to the receiving port through which it is connected to its parent. Each detected brick is consequently represented by a CubeNode containing:

- The complete node identifier
- The relative orientation
- The receiving port through which it is connected to its parent
- The child mask
- The subordinate nodes

After reconstruction, the parser verifies that the complete payload has been consumed. Incomplete node data or remaining bytes cause the payload to be rejected as invalid.

The target construction uses a separate but structurally equivalent tree representation. It is stored as a JSON object and converted by the backend into a TargetNode. Each target node contains the required brick type and orientation, while its child nodes are assigned to the expected receiving ports. Since physical bricks of the same type are interchangeable within the task, the target structure does not contain individual brick identifiers. Both representations nevertheless share the same hierarchical organisation and can therefore be compared recursively.

6.4.3 Target Comparison and Feedback Data Generation

Once the detected payload has been reconstructed, the comparison begins at the root nodes of the detected and target structures and proceeds recursively through the receiving ports of each brick.

For every position occurring in either structure, the algorithm compares the presence, type, relative orientation and structural placement of the corresponding brick. The individual node identifier is not used to determine whether a brick is correctly positioned but is retained to address the physical brick when generating local feedback.

The comparison then creates an evaluated tree that combines actual and expected information. Each position is assigned to one of five internal states, as summarised in Table 21.

State	Assignment by the comparison algorithm
CORRECT	A brick is present with the required type, orientation and connection
WRONG	A brick is present but differs from the target
ATTACHABLE	A target brick is missing, but its parent has already been placed correctly
BLOCKED	A target brick is missing and cannot yet be placed because its parent is missing or incorrect
UNEXPECTED	A brick is present where the target does not contain a brick

Table 21: *Internal states of the evaluated construction tree*

If a brick is connected to a receiving port that is not occupied in the target structure, the algorithm searches the remaining unoccupied ports of the same parent for an expected brick of the same type. If such a position exists, the brick is classified as WRONG and the expected port is stored in the comparison result. Otherwise, the brick and its subordinate structure are classified as UNEXPECTED. The comparison also considers the validity of the complete structural path. If a parent brick is missing or incorrect, its subordinate bricks cannot yet occupy valid positions within the target structure and are therefore marked as WRONG. Conversely, an unoccupied target position is classified as ATTACHABLE only if its parent path is correct.

In addition to these general internal states, each evaluated node also contains an error mask. Its individual bits indicate whether the deviation relates to the brick type, orientation, receiving port, parent path or the presence of an unexpected element. This additional separation allows the same comparison result to support different forms and levels of feedback.

When the comparison is complete, two output representations are generated from the evaluated tree: For the physical construction, a brick-specific feedback data set for all physically present nodes is generated and converted into the feedback frame defined in Section 6.2.3.2.1. For the frontend, the evaluated tree is converted into a JSON-compatible representation. This representation contains both detected bricks and currently unoccupied target positions, together with their assigned states and expected properties.

6.5 Frontend, Visualisation and Feedback

The backend transfers the evaluated tree to the frontend as a JSON object. Based on the structural and feedback information contained in this representation, the frontend calculates the spatial poses of the individual bricks and generates the user-facing visualisation described in the following section.

6.5.1 Calculation of Spatial Poses

The evaluated tree received from the backend contains no absolute spatial coordinates. Instead, the frontend calculates the position and orientation of each brick from its type, receiving port, relative orientation and parent–child relationship. The same calculation is used for target structures and evaluated constructions.

For the calculation, the implementation uses a local coordinate system for every brick: The z-axis represents the vertical axis of the brick, and the docking face is oriented towards its local negative z-direction. For the cube brick, receiving ports (0) to (4) correspond to the directions (-x), (-y), (+z), (+x) and (+y). The smashed brick provides only one receiving port in its local (+z)-direction.

For every connection, the local position of a child brick is determined by the direction and distance assigned to the receiving port of its parent:

$$\boldsymbol{p}_{\text{local}} = \boldsymbol{d}_{t,p} \times l_{t,p}$$

where t denotes the type of the parent brick, p the receiving port, $\boldsymbol{d}_{t,p}$ its direction vector and $l_{t,p}$ the distance between the model origins. The distance $l_{t,p}$ depends on the geometry of the parent brick and the selected receiving port. The reduced height of the smashed brick is therefore considered when positioning a child connected to its upper receiving face.

The local orientation of a child is represented using a quaternion that consists of an attachment rotation and an additional twist rotation:

$$q_{\text{local}} = q_{\text{attachment}}(t, p) \times q_{\text{twist}}(o).$$

The attachment quaternion $q_{\text{attachment}}(t, p)$ aligns the docking face of the child with the selected receiving face of the parent. The twist quaternion $q_{\text{twist}}(o)$ represents the detected relative rotation o. The four transmitted orientation values are mapped to rotations of 90°, 0°, −90° and −180° around the local z-axis. This assignment aligns the orientation of the digital models with the orientation convention used by the connection-detection system.

The first brick connected to the baseplate forms the root of the displayed construction and therefore has no parent brick. Because of this, its local position is set to the origin of the scene and its local orientation is derived solely from its transmitted relative rotation value. For every other subordinate brick, the local transformation is combined with the transformation of its parent:

$$T_{\text{global},c} = T_{\text{global},p} \times T_{\text{local},c}.$$

This calculation is applied recursively throughout the tree. Local connection information is therefore sufficient to derive the global position and orientation of every brick, and the backend does not need to transmit absolute coordinates.

Three-Dimensional Visualisation

The resulting construction is then displayed using corresponding GLB models of the two brick types placed within a Three.js scene. Using OrbitControls, the scene can be rotated and enlarged through mouse or touch input, allowing the construction to be inspected from different perspectives.

Figure 21: Correspondence between a physical brick construction and its reconstructed three-dimensional model

6.5.2 Digital and Local Feedback

The three-dimensional representation of the evaluated construction is augmented with visual cues derived from the comparison states described in Section 6.4. These cues are complemented by local

feedback from the LEDs integrated into the physical bricks. Table Table 22 summarises how each state is represented across both feedback levels, while Figure 22 illustrates the digital representation of the states CORRECT, WRONG and ATTACHABLE.

Evaluation state	Representation in the frontend	Feedback on the physical brick
Correct	Green pulsing outline	Green LED light
Wrong or unexpected	Red pulsing outline	Red LED light
Attachable	Semi-transparent ghost brick	No physical output
Blocked	Not displayed	No physical output

Table 22: *Implemented presentation of the evaluation states*

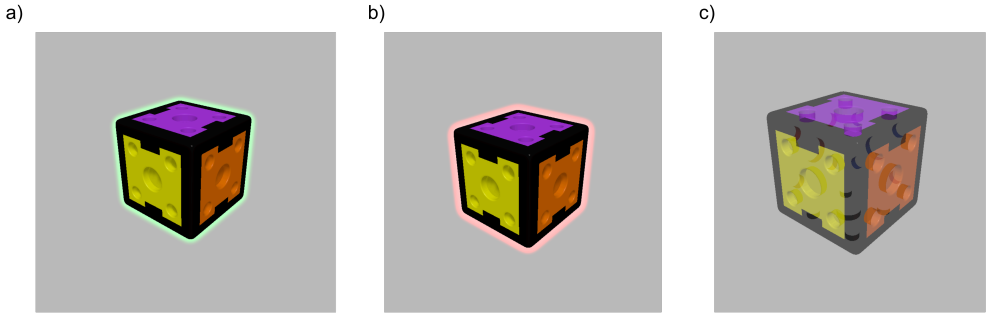

Figure 22: Implemented visual feedback for the states (a) CORRECT, (b) WRONG and (c) ATTACHABLE

Physically present bricks retain the appearance of their corresponding digital models, while coloured outlines indicate the evaluation result. Green represents a correctly positioned brick, whereas red indicates an incorrect or unexpected brick. A currently attachable target brick is displayed semi-transparently as a visual indication of an available next construction step. Target bricks that cannot yet be reached because their preceding construction path is incomplete are not displayed.

The global feedback in the frontend is coordinated with the local LED feedback of the physical bricks. A brick shown with a green or red outline is illuminated in the corresponding colour. Missing target bricks have no physical counterpart and are therefore represented exclusively in the frontend.

To indicate both the current construction progress and the completion of the task, a progress bar additionally shows the proportion of the target construction that has already been completed correctly.

6.5.3 User Interface and Task Selection

The visualisation process is not limited to displaying the detected bricks and their feedback states during the construction task. It begins on the dashboard, where the learner selects the target structure that defines the subsequent task.

The dashboard presents the available target constructions as selectable cards. Each card contains a three-dimensional preview generated from the corresponding target-tree data. Selecting a card opens an enlarged preview window in which the construction can be rotated and inspected from different viewing angles. The learner can then either cancel the selection and return to the dashboard or confirm the selected construction.

The preview window additionally contains two level options from which the learner can choose. Although the selected level is transferred together with the target structure, it does not currently affect the task behaviour. It is intended to support a future differentiation of task difficulty.

After the target structure has been confirmed, the live construction view is opened and the selected target is transferred to the backend. During the task, this view displays the evaluated construction using the spatial reconstruction and feedback representations described in the preceding sections. The progress bar is placed at the top of the interface and indicates the proportion of the target construction that has already been completed correctly.

The construction session can be terminated using the button in the upper area of the live view. The frontend then returns to the dashboard, where another target construction can be selected. The complete interaction sequence is illustrated in Figure 23.

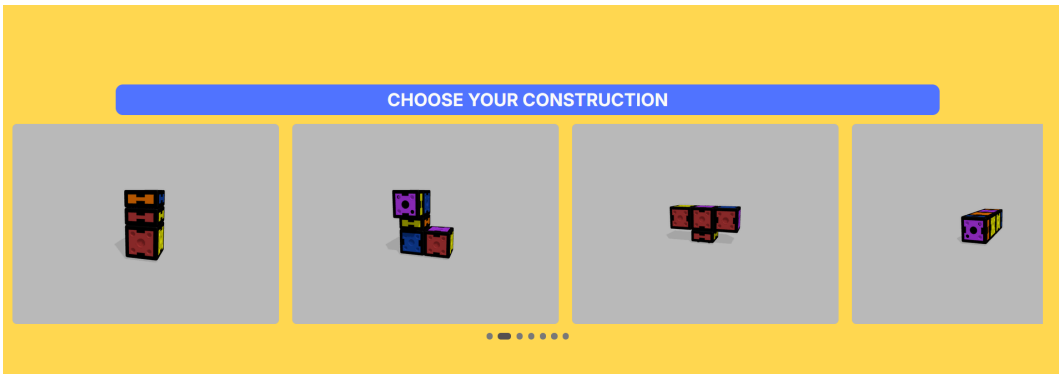

Figure 23: Target-selection dashboard of the learning application

7 Evaluation

The aim of the evaluation is to determine to what extent the developed prototype fulfils the requirements established in Section 4. Since the requirements concern different aspects, different evaluation methods were applied. These comprise a formative target-group evaluation and an analytical technical evaluation. Table 23 provides an overview of the applied evaluation methods and the requirements addressed by each method. Some requirements are associated with more than one method because individual aspects could be examined separately even when the complete integrated function could not be verified. At the time of the evaluation, the electronic construction-detection system was not sufficiently reliable. The complete detection and feedback process could therefore not be evaluated as an integrated system. The evaluation did not investigate long-term use or learning outcomes.

Evaluation method	Assessment procedure	Requirements addressed
Target-group evaluation	Observation of brick handling, usability, visual comprehension and assistance needs	TASK1; IA1–IA5; TUI1–TUI3; C1–C4
Physical inspection and demonstration	Inspection, dimensional checks, component count and manual demonstration of connections and assembly	G1; C1–C4; EL1–EL2; S1–S3
Implementation analysis	Traceability analysis of contact design, communication protocol, firmware and application components	DET1–DET7; FB1–FB4; IA6; EL3; S4
Cost analysis	Bill-of-materials calculation	EC1
Integrated-system test	Not conducted due to insufficient electronic reliability	DET1–DET7; FB1–FB2; IA6

Table 23: *Overview of the applied evaluation methods and corresponding requirements*

Assessment Criteria

Based on the available evidence, each requirement is classified as fulfilled, partially fulfilled, implemented but not verified, not fulfilled or not evaluated. A requirement is classified as fulfilled only if its relevant functionality or property could be demonstrated through the applied evaluation method. Requirements that were represented in the design or implementation but could not be tested in operation were classified as implemented but not verified.

7.1 Target-Group Evaluation Methodology

The target-group component was conducted as a formative qualitative evaluation to investigate the accessibility of the physical building bricks and the usability of the learning application for the intended target group. At the time of the target-group evaluation, the electronic construction-detection system was not yet sufficiently reliable for use during the construction tasks. The evaluation therefore focused on physical interaction with the building bricks and the use of the learning application, including the selection and visualisation of target structures and the learners' ability to relate the digital representations to the physical bricks.

Participants and Setting

The evaluation was conducted with 10 students attending the sixth year of a special education school. The group was a mix of students with different educational support needs related to **GE**, **LE** and **ASD**. The evaluation took place in the students' regular school environment and lasted approximately two hours. The tasks were initially completed individually, but during the later part

of the session, several students also worked together in pairs. The construction activities were supervised by the researcher and the attending teachers, who provided assistance when necessary.

Materials and Tasks

For the construction tasks, eight physical prototype bricks were available, comprising four bricks of each of the two implemented brick types. The electronic components were not used during the session. The learning application was presented on a touchscreen tablet and displayed a selection of target structures with varying levels of difficulty and complexity from which the students could choose. After confirming their selection, the chosen target structure was displayed as an interactive three-dimensional model in full-screen mode.

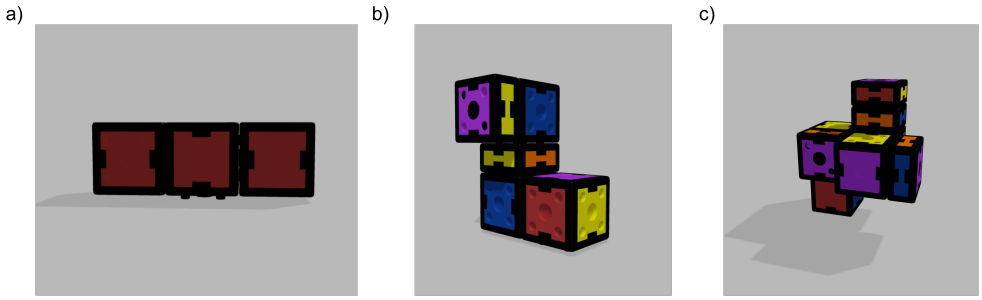

Figure 24: Examples of target structures used in the target-group evaluation: (a) low, (b) medium and (c) high structural complexity

Procedure

At the beginning of the evaluation, the students were introduced to the general purpose of the system and the reconstruction task. Each student was subsequently invited to select a target structure in the learning application and reproduce it using the physical bricks. Depending on their interest, the students could complete more than one task or work collaboratively in pairs, as shown in Figure 25. During construction, the students could interact with the three-dimensional visualisation of the target structure and rotate it to inspect the construction from different viewing angles. The students were encouraged to complete the tasks independently and without assistance from the supervising adults, but support was provided gradually if required. Initially, verbal hints were given. If these were insufficient, the target visualisation was rotated to provide a more suitable viewing angle and the relevant brick was indicated on the screen. If further assistance was required, the appropriate physical brick was provided to the student. Only when these forms of support proved insufficient did a supervising adult demonstrate or perform the required connection.

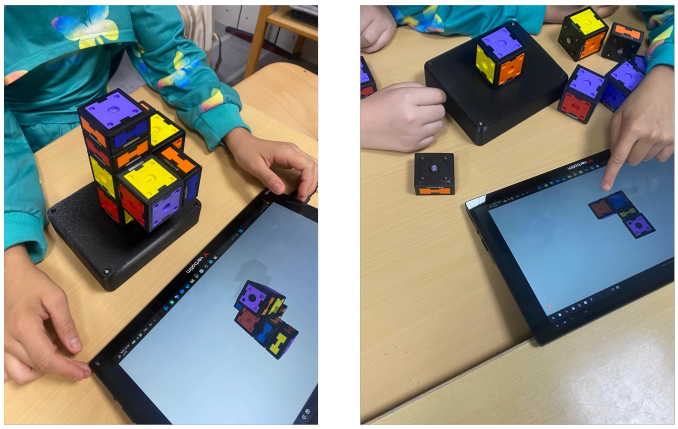

Figure 25: Learners during the construction process

7.1.1 Data Collection

Data were collected through qualitative observation of the students' interaction with the physical bricks and the learning application. The evaluation did not aim to measure learning outcomes. Instead, it examined the comprehensibility and usability of the interaction concept during the session.

Regarding the physical bricks, the observation focused on:

- whether the students recognised the objects as building bricks intended to be connected
- whether they understood which faces could be connected and how the two brick types differed
- whether they could grasp, position, connect and disconnect the bricks
- whether the magnetic connections provided sufficient stability while remaining easy to separate
- whether the students could use the bricks to construct three-dimensional arrangements

Regarding the learning application, the observation focused on:

- whether the students understood the construction task presented by the application
- whether they could relate the digital representations to the corresponding physical bricks
- whether they could select and confirm a target structure
- whether buttons and other interactive elements were recognised and operated correctly
- whether they could manipulate the three-dimensional model
- what type and level of assistance they required during these interactions

The observations were recorded in notes during and immediately after the session.

7.2 Results

The following sections focus on the findings that are most relevant to the central interaction concept and to the technical feasibility of the prototype. Requirements that produced similar or directly observable results are not discussed individually in the text in order to avoid repetition. A compact summary is provided at the end of the results, while the complete requirement-by-requirement assessment is included in Appendix Section A.

7.2.1 Target-Group Evaluation

Physical Interaction with the Building Bricks

The students immediately recognised the objects as building bricks and were able to grasp, position and connect them, including with one hand. The form-fit elements guided the connection, while the magnets assisted alignment and reduced the required precision. The construction activity therefore fulfilled the central physical interaction described by [TASK1] , [IA1] and [C3] . Connections could also be released and repositioned, allowing previous construction steps to be corrected as required by [IA5] .

After a brief explanation, the students generally understood which faces could be connected. Nevertheless, docking-face-to-docking-face combinations were attempted because the central magnets still produced a slight attraction even though the form-fit elements prevented complete engagement. The overall purpose of both brick types and their correspondence with the digital models were understood, but the connection rules were not entirely self-explanatory. The physical aspect of [TUI3] was therefore only partially fulfilled.

The bricks enabled three-dimensional arrangements and different relative orientations, providing evidence for [C1] and [C2] . Their stability was sufficient for smaller constructions, but horizontal sequences of more than three bricks separated under their own load. Requirement [C4] was consequently only partially fulfilled: the connections required little force to create and release but did not support every tested arrangement reliably.

Learning Application and Visual Comprehension

The students were generally able to browse the available target structures independently and recognised differences in complexity. The previews were usually viewed only briefly and did not contribute substantially to the selection. Confirming a target frequently required assistance because the corresponding button was comparatively small and did not respond reliably. Although the remaining touch interaction was understood, these difficulties and the limited scope of the tested application mean that [IA2] was only partially fulfilled.

The students understood that the full-screen model could be rotated and zoomed and regularly used these functions when the arrangement was unclear. However, some had difficulty distinguishing the two digital brick types, identifying the root brick and determining the orientation of a child relative to its parent. Simple target structures were easier to interpret than complex ones, and some students required an initial demonstration before continuing independently. The clarity of the task required by [IA3] and the consistency of the physical–digital mapping required by [TUI3] were therefore only partially achieved.

The available targets varied in the number, arrangement and orientation of their bricks, thereby implementing the basic variation of difficulty required by [TUI2] . Students were free to select targets, try different construction steps and correct unsuccessful attempts. During pair work, students who had already understood the task also provided hints to their partners. These observations support the structured openness described by [TUI1] , although the extent of independent use differed between participants.

7.2.2 Analytical Prototype Assessment

Construction Detection and Data Processing

The implementation analysis showed that the data format and firmware contain the information required for unique brick identification, connection-face assignment and relative orientation. The node identifier combines brick type and instance, occupied receiving faces are represented through a child mask, and one of four upstream contacts determines the relative rotation. The mechanisms required by [DET1] , [DET2] and [DET3] are therefore implemented, but their operation could not be verified in the integrated electronic system.

The recursive communication and tree representation are designed to support constructions containing multiple bricks. Since reliable simultaneous detection was not demonstrated, [DET5] remains not evaluated. Frame-length checks, checksums, timeouts and parser validation provide error handling for invalid communication data, partially fulfilling [DET6] ambiguous physical contact states were not examined.

The response-time requirement was not met. The firmware inserts a delay of 550 ms between detection cycles, while [DET7] specifies a maximum of 400 ms for detecting a change. Communication, retries and subsequent processing add further time, so [DET7] is classified as not fulfilled even without an end-to-end measurement.

Target Comparison and Feedback

The backend implements recursive comparison of actual and target trees by brick type, receiving face, orientation and structural path. It derives evaluated node states and generates both a frontend representation and identifier-specific LED data. This provides an implementation of [DET4] , [FB1] and [FB2] , but the complete path from a detected physical construction to coordinated visual and local feedback was not verified. These requirements are therefore classified as implemented but not verified.

Graduated support is not yet operational. Although the comparison distinguishes several error states and the interface includes level options, the selected level does not change the task presentation, feedback or assistance. Requirements [FB3] and [FB4] are consequently not fulfilled.

The prototype combines intrinsic haptic feedback with visual output in the application and on the bricks, but coordinated system-generated feedback was not tested and remains primarily visual. Requirement [IA6] is therefore only partially fulfilled.

Further Prototype Constraints

Direct inspection confirmed the implementation of two geometrically distinct brick types, a central power supply without individual batteries and a housing that can be assembled and disassembled. However, only eight of the ten bricks required by [EL2] were produced, and the material cost of approximately EUR 5 per brick exceeded the EUR 3 target specified by [EC1] . Both requirements are therefore not fulfilled.

No damage occurred during ordinary construction activities, but no standardised durability test was conducted. One brick broke after being thrown forcefully onto the floor, demonstrating limited impact resistance beyond the intended handling conditions. The recessed contacts, mechanical guidance and open-drain-like communication reduce the risk of incorrect contact and electrical contention, but no dedicated overvoltage test or protection verification was performed. The relevant durability and electrical-safety requirements are therefore assessed conservatively in the complete overview table.

7.3 Overall Requirement Fulfilment

Of the 34 requirements, 12 were fulfilled, 10 were partially fulfilled, six were implemented but could not be verified, five were not fulfilled and one could not be evaluated. Table Table 24 summarises these results. The complete evidence for each requirement is documented in Appendix Section A.

Assessment	Number	Requirements
Fulfilled	12	TASK1; IA1; IA4–IA5; TUI1–TUI2; G1; C1–C3; EL1; S3
Partially fulfilled	10	IA2–IA3; TUI3; IA6; C4; DET6; EL3; S1–S2; S4
Implemented but not verified	6	FB1–FB2; DET1–DET4
Not fulfilled	5	FB3–FB4; DET7; EL2; EC1
Not evaluated	1	DET5

Table 24: *Summary of requirement fulfilment*

7.4 Discussion

7.4.1 Interpretation of the Results

The objective of this work was to develop a tangible learning game that enriches classroom activities for children in special education and supports teachers in providing action-oriented learning opportunities. The reconstruction tasks were designed primarily to address spatial thinking, action planning and self-directed learning. They also involve problem-solving, error recognition and self-correction, while the physical interaction additionally engages fine motor skills, concentration and hand–eye coordination.

The evaluation indicates that the prototype successfully translates this objective into an accessible and motivating form of interaction. The students understood the central idea of reproducing a digital target with physical bricks and engaged willingly with the construction activity. They selected structures, manipulated the three-dimensional representation and built, inspected and modified tangible results. It is particularly encouraging that the learning concept remained comprehensible and engaging even without automatic detection and feedback. This suggests that the central value of the system is not created by technology alone, but by the coherent combination of a meaningful task, a digital reference and an actively constructed physical artefact.

This combination is closely aligned with the constructionist principle of *learning by making* discussed in Section 2.1.3. Instead of receiving a completed solution, learners externalise their understanding in a construction that remains visible and can be changed through further action. Experimentation and correction are therefore part of the activity rather than signs that the task has failed. The observed willingness to try different approaches supports the view that the prototype creates a suitable basis for exploratory learning. At the same time, the physical positioning and rotation of the bricks correspond directly to the spatial relationships to be understood. The interaction thus reflects the principles of *embodied cognition* and *embodied interaction* described in Section 2.2.1: bodily action is meaningfully connected to the subject matter rather than merely added as an arbitrary activity.

The prototype also responds convincingly to the design principles for child-appropriate TUIs presented in Section 2.6. The target structure provides a clear framework, while the construction process leaves room for different sequences, repeated attempts and individual strategies. Structures of varying complexity allowed students with different support needs to participate within the same interaction concept. This balance between guidance and openness is particularly relevant for the heterogeneous learning prerequisites described in Section 1.1. Moreover, the physical availability of several bricks supported spontaneous cooperation: students exchanged suggestions and more confident learners assisted their peers. The system can therefore encourage not only individual construction but also socially shared learning without requiring a separate collaborative mode.

Although differentiated assistance, multimodal output and the complete feedback process remain to be implemented reliably, the prototype already establishes the conceptual and technical foundation for these extensions. The evaluation therefore does not merely show that individual components are usable; it demonstrates that the overall learning idea is understandable, engaging and appropriate for the intended context. The prototype can already enrich teacher-guided activities and offers a credible path towards more independent practice once automatic feedback is fully operational. While the present study does not establish measurable learning gains, it provides strong formative evidence that the system is worth developing further as an educational TUI for special education.

7.4.2 Limitations

The limitations concern both the available evidence and the current state of the prototype. The target-group evaluation involved ten students from one class at a single special education school. Although different support needs were represented, the group cannot reflect the full heterogeneity of learners with needs related to **GE**, **LE** and **ASD**. Moreover, the evaluation consisted of one two-hour session and relied on qualitative observation notes. No standardised usability measures, systematic coding scheme or independent observation were used, and the assistance provided varied according to individual need. The findings therefore offer formative insight into the interaction concept but do not permit representative comparisons between learners or broader generalisation.

The most important functional limitation is that the children did not experience the complete tangible user interface. Because electronic construction detection was not sufficiently reliable, their actions were neither automatically reconstructed nor compared with the target during the session. They consequently could not test whether the system distinguishes technical detection problems from construction errors, whether feedback is understandable and timely, or whether it enables them to identify and correct errors without adult support. The implementation analysis confirms that the intended processing and feedback mechanisms are conceptually represented, but not that they operate reliably and comprehensibly in actual learning situations.

Individual adaptation is also incomplete. Difficulty can currently be varied through the target structures, but the selected support level does not yet change the information or assistance provided. In particular, no graduated feedback is available that begins with a simple indication

of an error and adds explanations or suggestions only when required. The children therefore did not test the stepwise feedback concept. This is especially relevant for the heterogeneous target group: some learners may need only confirmation, whereas others require additional orientation to continue independently. Until these levels are implemented and evaluated, the central objective of supporting self-directed work can only be considered partially demonstrated.

Accessibility is further limited because the application and the planned system-generated feedback are predominantly visual. Auditory alternatives, such as spoken instructions or read-aloud assistance, are not yet available, and the suitability of colour contrast, text quantity and pictorial guidance for different perceptual and cognitive needs has not been systematically assessed. The current prototype therefore does not yet provide the adaptable, multimodal access required to accommodate the full range of intended users.

Finally, the evaluation examined immediate interaction rather than educational effectiveness or sustained classroom use. It did not measure learning gains, transfer, concentration or changes in independent task performance across repeated sessions. Teachers were present during the activity, but effects on preparation effort, classroom integration and the need for supervision were not evaluated systematically. The limited number of bricks and the focus on one reconstruction-task format further restricted the possible activities, while durability and electrical safety were assessed only informally. These limitations do not call the underlying learning concept into question, but they define the remaining work required before the prototype can fulfil its intended educational role as a reliable and individually accessible classroom system.

8 Conclusion

Summarise the work and provide a final conclusion in approximately half a page.

8.1 Outlook

Make the prototype fully functional and test it again with the students at the school. Extend the learning application with different feedback levels. Add different learning games, such as having one student construct a model for another student to reproduce. Investigate replacing the computer with an ESP32 and a Wi-Fi connection to the tablet. Make the feedback in the learning application multimodal by supplementing text output with spoken feedback, for example.

Methodology: What was evaluated, how, with whom and according to which criteria? Results: What was observed or determined? Discussion: What do these results mean?

8.2 Notes

- Students helped one another; space multiplexing automatically encouraged collaborative work.
- The students experimented extensively and were not afraid of making mistakes.

From the conversation with the teacher:

- Make the learning application multimodal so that feedback can be heard as well as seen, for example by reading assistance text aloud.
- Use more contrasting colours for the bricks.
- Use many images and little text in the learning application.

The system-generated feedback is primarily visual and therefore does not constitute fully multidimensional feedback in the sense of IA6. It is nevertheless distributed across several complementary levels.

- The electronics, including detection of the current construction, its visualisation, comparison with the target construction and the resulting feedback, were not used. Consequently, the

evaluation covered only the physical bricks and the learning application. For the physical bricks, it examined whether the students could grasp, connect and separate them, whether they understood which faces could and could not be connected, whether they recognised the objects as connectable bricks, and whether the balance between magnetic stability and ease of connection was appropriate. For the learning application, it examined the selection and presentation of a complete target construction, including whether the target selection was comprehensible, whether the students could determine what they needed to build from the visualised target and whether they could relate the physical bricks to their digital representations.

- Some students immediately understood that they had to reproduce the target construction with the bricks. Others first required several demonstrations of how to connect the bricks before continuing independently.

9 List of Abbreviations

3D	Three-Dimensional	**NFC**	Near Field Communication
ASD	Autism Spectrum Disorder	**RFID**	Radio Frequency Identification
GE	Special education focus on intellectual development	**TUI**	Tangible User Interface
GUI	Graphical User Interface	**UHF**	Ultra High Frequency
ID	Identification	**UI**	User Interface
LE	Special education focus on learning	**rep_d**	Digital Representation
LED	Light-Emitting Diode	**rep_p**	Physical Representation
MVC	Model View Controller		

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A Detailed Requirement Assessment

The following table provides the complete requirement-by-requirement assessment of the prototype. The classifications refer to its state at the time of evaluation and consolidate evidence from the target-group evaluation, physical inspection and demonstration, and implementation analysis.

ID	Evidence	Assessment
TASK1	Active construction with physical bricks was demonstrated	Fulfilled
IA1	Constructions remained visible, inspectable and modifiable	Fulfilled
IA2	Basic use was understood, but confirmation frequently required assistance	Partially fulfilled
IA3	Root, brick type and orientation were not always understood	Partially fulfilled
IA4	Tasks imposed no time limit and permitted individual pacing	Fulfilled
IA5	Connections could be corrected, repeated and repositioned	Fulfilled
TUI1	Target-based framework allowed experimentation and different construction sequences	Fulfilled
TUI2	Targets varied in brick count, arrangement and orientation	Fulfilled
TUI3	General physical–digital correspondence was understood, but some cues remained ambiguous	Partially fulfilled
FB1	Visual and local feedback processing exists but was not tested end to end	Implemented but not verified
FB2	Feedback states are assigned to affected bricks and positions but were not user-tested	Implemented but not verified
FB3	No progressive increase of feedback after unsuccessful correction	Not fulfilled
FB4	Level selection does not alter assistance or task behaviour	Not fulfilled
IA6	Haptic and visual levels exist, but coordinated feedback remains primarily visual	Partially fulfilled
G1	Cube and smashed brick were implemented as distinct geometries	Fulfilled
C1	Three-dimensional and branching arrangements were demonstrated	Fulfilled
C2	Connections in different relative orientations were demonstrated	Fulfilled
C3	One-handed grasping, positioning and connection were observed	Fulfilled
C4	Connections were easy to operate but unstable in longer horizontal arrangements	Partially fulfilled

ID	Evidence	Assessment
DET1	Type and instance are encoded in the node identifier	Implemented but not verified
DET2	Child mask and port order encode connection relationships	Implemented but not verified
DET3	Four upstream contacts encode relative orientation	Implemented but not verified
DET4	Recursive actual–target comparison is implemented in the backend	Implemented but not verified
DET5	Reliable electronic operation with multiple bricks was not demonstrated	Not evaluated
DET6	Invalid frames and payloads are rejected; ambiguous physical states were not tested	Partially fulfilled
DET7	A 550 ms cycle delay already exceeds the specified 400 ms maximum	Not fulfilled
EL1	Bricks contain no individual batteries and use a central supply	Fulfilled
EL2	Eight bricks were produced instead of the required ten	Not fulfilled
EL3	Recursive operation supports extension, but no expansion test was conducted	Partially fulfilled
S1	Ordinary use caused no damage, but no standardised durability test was conducted	Partially fulfilled
S2	The intact housing encloses components, but protection under impact is limited	Partially fulfilled
S3	Assembly and disassembly were demonstrated without damage	Fulfilled
S4	The interface reduces short-circuit risk, but overvoltage protection was not verified	Partially fulfilled
EC1	Material cost was approximately EUR 5 instead of the EUR 3 target	Not fulfilled

Table 25: *Detailed assessment of requirement fulfilment*